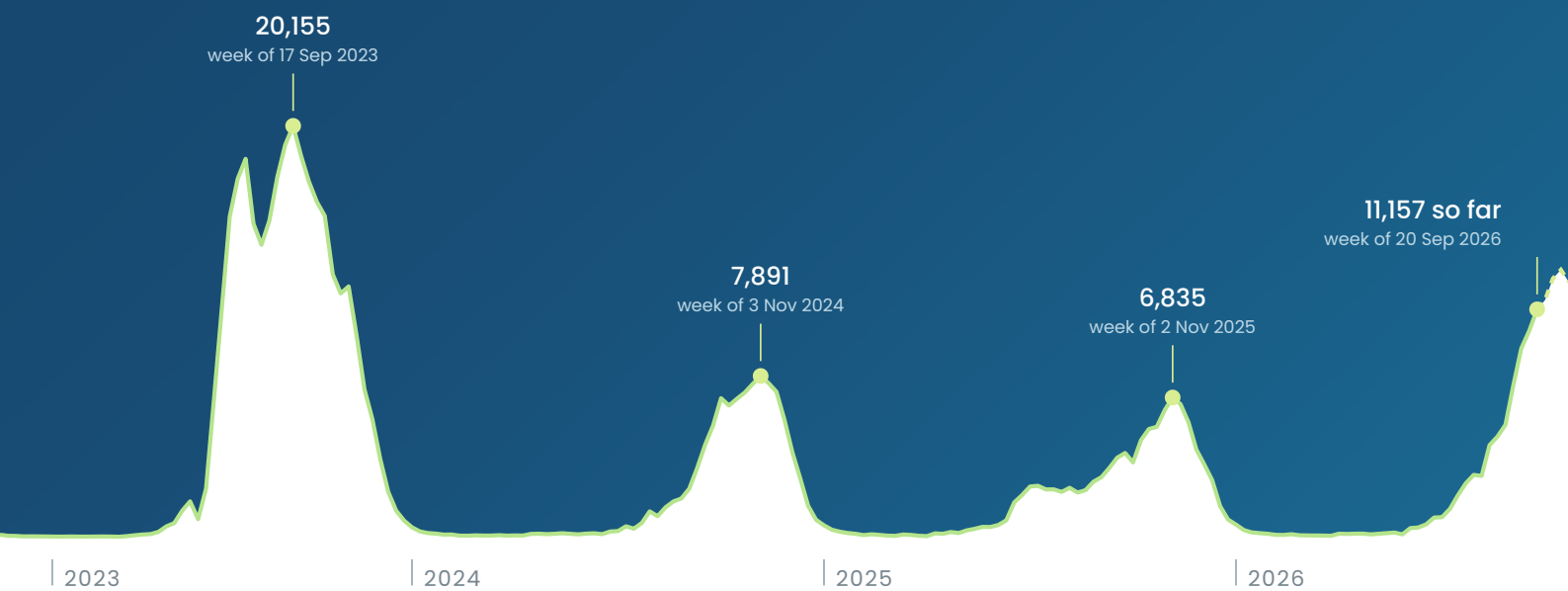


Forecasting Dengue Hospital Admissions in Bangladesh

Weekly forecasts one to four weeks ahead for the eight divisions, built on DGHS surveillance data from 2023 to 2026



Weekly dengue hospital admissions in Bangladesh, January 2023 to September 2026

Dashed: four-week forecast

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DATA

DGHS Dengue Dynamic Dashboard
Open-Meteo weather archive (ERA5)
BBS Population Census 2022

METHODS

Python and DuckDB SQL
R: STL, DLNM, ARIMA, ETS
XGBoost and random forest
Power BI dashboard

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Executive summary

Dengue fills hospital wards in Bangladesh every year, and the size and timing of the season change from one year to the next. This project asks how many dengue admissions each division should expect one to four weeks ahead, and how far such a forecast can be trusted.

592,614

hospital admissions in the data, January 2023 to 20 September 2026

2,912

deaths in the same period

1,560

division-weeks: 8 divisions over 195 weeks

8

forecasts compared on the same weeks of 2025 and 2026

What was done. The daily and weekly counts behind the DGHS dengue dashboard were collected, checked against each other and repaired, most of all the 2023 division series, where about 105,000 admissions were missing and were rebuilt from the daily press releases. The cleaned data was analysed with SQL, exploratory figures, a seasonal decomposition and a distributed lag model of weather. Five forecasting models, an ensemble of them and two forecasts that need no model were then scored on every week of 2025 and 2026. All results are shown in an eight-page Power BI dashboard.

Main findings.

1. **Burden.** 2023 had 316,906 admissions and 1,703 deaths. 2024 and 2025 had about 101,000 and 103,000 admissions. By week 39, 2026 had 71,949, which is 1.58 times 2025 at the same week.
2. **Season.** One yearly cycle dominates (seasonal strength 0.82). The season began between mid June and late July in each year, and peaked in mid September in 2023 but in early November in 2024 and 2025.
3. **Geography.** Barishal has the highest admission rate per person in three of the four years. Dhaka division's share of national admissions fell from 53.5% in 2023 to 40.3% in 2026 so far. The other divisions do not run steadily behind Dhaka.
4. **People.** Ages 16 to 30 are 40% of admissions and 61% of admissions are male, but women are 48% to 57% of deaths. Deaths per 100 admissions rise steeply with age.
5. **Weather.** Raw correlations of 0.5 to 0.8 between weather and later admissions fall to about 0.1 once the shared season is removed. In the main model, cooler and drier spells than usual are followed by fewer admissions (relative risk 0.23 and 0.64). Hot and wet spells show no clear effect.
6. **Forecasts.** No model was best in both years. XGBoost led in 2025 (mean absolute error 53.1 admissions per division and week) and ARIMA with Fourier terms led in 2026 (44.3). The median of the five models was third and then second, and had the lowest error over the two years together. Repeating the latest week gave 90.6 and 96.5.
7. **Trust.** Only XGBoost was clearly better than repeating the latest week in both years by the Diebold-Mariano test. The 80% forecast ranges held 81% to 92% of actual values. Weather features improved the 2025 forecasts by 5.9% and the 2026 forecasts by only 0.8%.

The next four weeks. From the week of 20 September 2026, which had 11,157 admissions, the ensemble expects 11,612, 12,673, 13,097 and 12,499 admissions in the following four weeks. The individual models spread from 9,161 to 15,529 by the fourth week.

How far to trust it. The models have seen three seasons, and the test year ends while cases are still rising. The numbers are hospital admissions reported by DGHS, not all infections, and part of the 2023 division series is an estimate. Section 13 lists the limits in full.

1 Introduction

This report documents a complete forecasting project on dengue in Bangladesh: how the data was collected and repaired, what four years of hospital admissions show, and how well eight different forecasts did when they were tested on weeks they had never seen.

1.1 Background

Dengue is a viral infection spread by *Aedes* mosquitoes. One widely cited estimate puts the number of infections at about 390 million a year worldwide [1]. In Bangladesh it returns every year with the monsoon, and the yearly burden has risen over the last two decades [2]. The outbreak of 2023 was the largest the country has recorded [3]. By the end of that year the Directorate General of Health Services (DGHS) had reported 321,179 hospital admissions and 1,705 deaths [4].

For a hospital the useful question is a practical one: how many dengue patients should we expect in the next few weeks? A forecast one to four weeks ahead is long enough to plan beds, staff and supplies, and short enough to be checked against what really happens.

DGHS publishes its daily counts on a public dashboard [4], but only as charts. There is no download button, and I could not find an open, cleaned, division-level weekly series for recent years. So the first half of this project is data work: collecting the numbers behind the charts, checking them against each other, and repairing the parts that do not add up. The second half is the analysis and the forecasts.

1.2 Objectives

The project has one main objective and seven public questions. The objective is to forecast weekly dengue hospital admissions in each of the eight divisions one to four weeks ahead, and to say honestly how far those forecasts can be trusted. The questions are the ones a reader outside statistics would ask, and every step of the analysis was planned against them.

Table 1. The seven questions of the project.

	Question	Answered in section
Q1	How large is the burden, and how does it change from year to year?	4, 5
Q2	When does the season start and peak, and how long does it last?	4, 6
Q3	Which divisions are hit hardest, and is dengue spreading beyond Dhaka?	4, 5, 6
Q4	Who is admitted and who dies, by age and sex?	5
Q5	Do rain, temperature and humidity lead admissions, and by how many weeks?	5, 7
Q6	How many admissions should hospitals expect one to four weeks ahead?	8, 9, 10
Q7	Does machine learning beat simpler models, and what drives the forecasts?	8, 10

1.3 Scope and definitions

- **Outcome.** People admitted to hospital with dengue, as reported by DGHS. This is a count of hospital cases. It is not the number of infections, most of which are never admitted.
- **Unit.** One division in one week. There are 8 divisions and 195 weeks, from the week of 1 January 2023 to the week of 20 September 2026, which gives 1,560 division-weeks.
- **Weeks and years.** Weeks run from Sunday to Saturday and are named by their Sunday. A week belongs to the epidemiological year of its Thursday, so yearly totals here differ a little from calendar-year totals.
- **Horizon.** Forecasts are made 1, 2, 3 and 4 weeks ahead.
- **Not covered.** District-level forecasts, years before 2023 (the dashboard does not return them), and causal claims about weather. The weather analysis describes associations only.

1.4 Workflow and evaluation design

The work is split across nine code files and one dashboard. Python handles collection, cleaning, SQL, exploration and the machine learning models. R handles the classical time series work, because the `forecast` and `dlnm` packages are more complete for it. The sections of this report follow the order in which the results build on each other, so the R scripts on seasonality and weather come before the forecasting notebook.

Table 2. The files of the project in the order they are reported.

File	Tool	What it does	Section
01_data_collection.ipynb	Python	Download the dashboard charts, weather and population; save dated raw files	2
02_data_audit_cleaning.ipynb	Python	Audit, reconcile and repair the series; one weekly calendar; train, validation and test split	3
03_duckdb_sql_analysis.ipynb	SQL in Python	Eleven DuckDB queries on burden, timing, geography, age and sex, weather and the no-model benchmark	4
04_exploratory_analysis.ipynb	Python	Figures and the checks a forecasting model depends on	5
06_seasonality.R	R	STL decomposition, season onset, peak and length	6
07_weather_dlnm.R	R	Distributed lag non-linear model of temperature and rainfall	7
05_ml_forecasts.ipynb	Python	XGBoost and random forest forecasts, ranges and SHAP	8
08_statistical_forecasts.R	R	ARIMA with Fourier terms (Box-Jenkins), ETS and a pooled negative binomial model	9
09_model_comparison_powerbi.R	R	All models on the same forecasts, Diebold-Mariano tests, ensemble, Power BI tables	10
Dengue Dashboard.pbix	Power BI	Eight report pages built on the exported tables	11

The years are used in a fixed way, set once in notebook 02 and read by every later file. Models learn from **2023 and 2024**. Every choice of features and settings is made on **2025**, the validation year. **2026** is the test year and is scored once, after all choices are fixed. Weeks are never shuffled into random groups, a forecast made in one week uses only what was known in that week, and every model is compared with two forecasts that need no model at all.

2 Data sources and collection

Notebook 01_data_collection.ipynb

Notebook 01 collects every raw input and saves it untouched, with the download date in the file name. Nothing is cleaned at this stage. The dashboard changes every day and can revise past numbers, so a dated snapshot is the only way to make each later result traceable. All results in this report rest on the snapshot of **1 October 2026**.

Table 3. Data sources.

Source	What was taken	Period	Used for
DGHS Dengue Dynamic Dashboard	Daily national admissions and deaths; weekly admissions by division; admissions and deaths by age group and sex	2023 to 1 Oct 2026	Main series
DGHS daily press releases (PDF)	Dhaka city and Barishal admissions per day, 9 Jul to 31 Dec 2023 (176 files)	2023	Repair of the 2023 division series
Open-Meteo historical weather API	Daily mean, maximum and minimum temperature, rainfall, relative humidity and dew point at the eight divisional headquarters (ERA5 reanalysis)	1 Sep 2022 to 24 Sep 2026	Weather analysis and optional model features
BBS Population and Housing Census 2022	Enumerated population by division, total 165,158,616	2022	Admissions per 100,000 and the model offset

2.1 DGHS dengue dashboard

The dashboard has no export function. Each chart is drawn with Highcharts, and its numbers sit inside the page as JavaScript arrays. The year filter is a form that sends `filter_year` by POST. The notebook requests each year, saves the raw page, and reads every chart's categories and series with a small bracket-matching parser, so nested objects inside a chart do not break the reading.

Three things were checked before relying on it. The years 2024 to 2026 are in the year dropdown, and 2023 is hidden in the page code but the server still returns it. The year 2022 and earlier come back empty, which is why the project starts in 2023. And each download is tested: the dates of the daily chart must end in the requested year, otherwise the row is marked `WRONG YEAR` and the run stops. All four years returned correctly, with 27 charts each (26 for 2025). A pause of three seconds between requests keeps the load on the DGHS server light.

Python Notebook 01: one request per year, raw page saved, and a check that the server returned the requested year (shortened)

```
DASH_URL = "https://dashboard.dghs.gov.bd/pages/heoc_dengue_v1.php"
YEARS = [2023, 2024, 2025, 2026]

for year in YEARS:
    r = requests.post(DASH_URL, data={"filter_year": year, "report_date_filter": "", "search_filter": "Search"}, headers=HEADERS, timeout=60)
    (DASH_DIR / f"dghs_dashboard_{year}_{RUN_DATE}.html").write_text(r.text, encoding="utf-8")
    charts = parse_charts(r.text)
    days = charts.get("confirmed_case", {}).get("categories", [])
    returned = sum(str(c).endswith(f"-{year % 100:02d}") for c in days) > 0.9 * max(len(days), 1)
    ...
    manifest.append({"source": "dghs_dashboard", "item": year, "status": "ok" if returned else "WRONG YEAR", "n_charts": len(charts), ...})
    time.sleep(3)
```

2.2 Weather

Daily weather comes from the Open-Meteo historical API [5], which serves the ERA5 reanalysis [6]. It is free, needs no key, and uses the same model for every division and every day. Station data from the Bangladesh Meteorological Department is not openly downloadable. Four variables are used: rainfall, which creates breeding sites; temperature, which controls mosquito survival and how fast the virus develops inside the mosquito [7]; and relative humidity and dew point as two measures of moisture.

Each division is represented by one point, the divisional headquarters city. That is a proxy for a whole division and is listed again under the limitations. The series starts on 1 September 2022, so that lags of up to 16 weeks exist for the first weeks of 2023.

2.3 Population

Population by division is taken from the enumerated counts of the Population and Housing Census 2022 [8]. It turns admissions into admissions per 100,000 people, so that a large division is not mistaken for a high-risk one. The same 2022 figure is used for all four years, which ignores population growth. Over four years that changes the rates only slightly, and it changes no comparison between divisions.

Table 4. The eight divisions, their census population and the point used for weather.

Division	Population, 2022 census	Share of population, %	Weather point, latitude	Longitude
Dhaka	44,215,107	26.8	23.81	90.41
Chattogram	33,202,326	20.1	22.36	91.78
Barishal	9,100,102	5.5	22.70	90.37
Khulna	17,416,645	10.5	22.85	89.54
Rajshahi	20,353,119	12.3	24.37	88.60
Mymensingh	12,225,498	7.4	24.75	90.41
Rangpur	17,610,956	10.7	25.74	89.28
Sylhet	11,034,863	6.7	24.89	91.87

3 Data audit and cleaning

Notebook 02_data_audit_cleaning.ipynb

This notebook turns the raw charts into clean, reconciled tables. It took more time than any model in the project, and it changed the results more than any model choice did. Every rule is logged with the number of cases it touched, and the raw files are never edited.

3.1 Year-specific charts and the daily series

When the dashboard is asked for a past year, some of its charts still show the current year. The proof is simple: the "admitted this year" card shows the same number in every year's download. Those charts were dropped. Eight charts really change with the year and were kept: daily admissions, daily deaths, weekly national counts, weekly counts by division, and admissions and deaths by age and by sex.

The daily series has 27 days with no report, 19 of them between February and April 2023 and 8 in 2024. They are kept as missing. A day with no report is not a day with no patients, so filling it with zero would be wrong. The table below compares the daily series with the dashboard's own yearly totals. In 2023 the daily series covers 98.75% of the yearly total, and the gap of 4,005 admissions is what was reported on the missing days. The other years agree to within 6 admissions.

Table 5. The daily series against the dashboard yearly totals (calendar years; 2026 up to 1 October).

Year	Days reported	Admissions in the daily series	Deaths	Dashboard's yearly total	Gap	Coverage, %
2023	346	317,012	1,705	321,017	4,005	98.75
2024	358	101,211	575	101,211	0	100.00
2025	365	102,855	413	102,861	6	99.99
2026	274	81,598	249	81,598	0	100.00

For 2023 the press release of 31 December 2023 reports 321,179 admissions, slightly above the dashboard figure, because the dashboard series was revised later. The dashboard series is used throughout.

3.2 One weekly calendar

The dashboard labels its weeks W01, W02 and so on, but the labels do not follow a fixed calendar. Checked against the daily data, the last label of 2024 covers ten days, the first label of 2026 covers 1 to 10 January, and in 2025 one week is split over two labels, so every later label sits one week earlier than its number suggests. A fixed rule from label number to date would put part of each year in the wrong week.

The labels were therefore matched to dates by the data. Labels stay in order, each goes to one Sunday week, and the arrangement chosen is the one where the label totals best match the daily data summed into Sunday weeks. Sharing or skipping a week carries a penalty of 1% of the year's cases, so it only happens where it clearly improves the fit. The error left after matching was 0.0% of cases in 2023, 0.69% in 2024, 3.09% in 2025 and 4.66% in 2026. That remainder comes from the dashboard placing a day or two on the wrong side of a week boundary, and it is corrected in section 3.4.

3.3 Repairing the 2023 division series

For 2024 to 2026 the division series adds up to the national series over the year. In 2023 it does not: about 105,000 admissions are missing. The weekly numbers show where. Up to the week of 2 July 2023 the divisions match the national total. Then the gap jumps, while the Dhaka series falls from 6,621 admissions in the week of 16 July to 2,772 in the week of 23 July, in the middle of the worst outbreak on record. At the same time Barishal reports exactly zero for three weeks in a row.

These are two separate problems. From late July 2023 the dashboard's Dhaka series counts only Dhaka division outside the two city corporations of Dhaka; from 2024 the city corporations are included again. And Barishal's numbers are missing or far too low for several weeks around its own peak.

The fix uses a second official source. The DGHS daily press releases were downloaded for all 176 days from 9 July to 31 December 2023. Two things are read from each PDF:

- **Dhaka city admissions**, from the summary table that splits the day's admissions into Dhaka city and outside Dhaka city. The day's national total from the dashboard is the anchor: the two numbers must add up exactly to it, which makes the reading self-checking. This number was read on 172 of the 176 days. The 4 days without it are 9 to 12 July, before the reporting change, so no value had to be estimated.
- **Barishal admissions**, from the division table. A row is accepted only if it passes two identities that every row of that table obeys (government plus private equals total new admissions, and cumulative cases minus deaths minus discharges equals patients still in hospital). A week uses the press-release value only if Barishal was accepted on every reported day of that week, which holds for 8 weeks from 16 July to 3 September.

In those 8 weeks Barishal takes the press-release value and Dhaka takes the rest of the gap. In the other weeks Dhaka city admissions are added to Dhaka from 23 July, and a leftover above 3% of the week's national total goes to Barishal, which happened in three weeks. The week of 3 September gives a direct check: there the dashboard's own Barishal value is intact, and it is 2,342 in both sources.

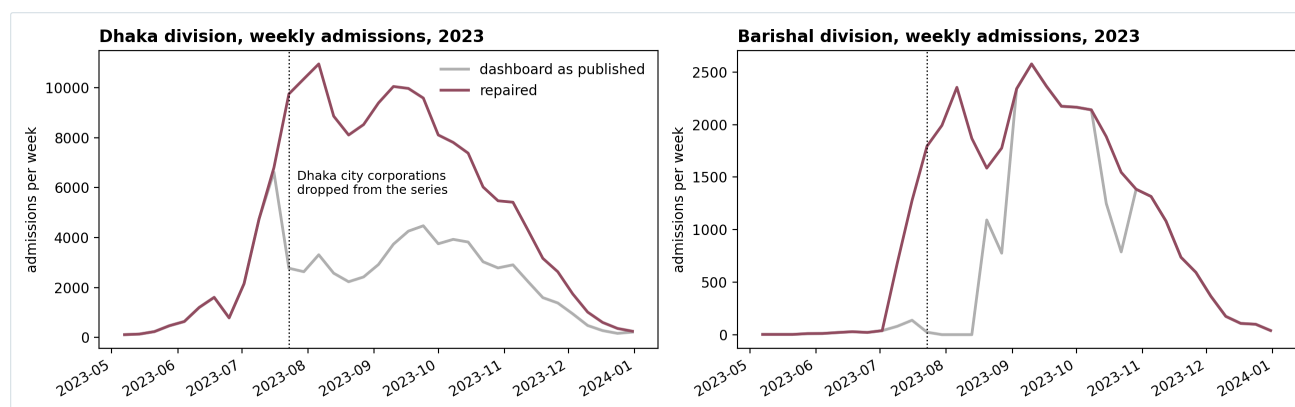


Figure 1. Dhaka and Barishal divisions in 2023, as published on the dashboard (grey) and after the repair. The dotted line marks the week from which the Dhaka series stopped including the city corporations.

Table 6. The 2023 repair, weeks of 9 July to 10 September. All values are weekly admissions.

Week of	National	Missing from divisions	Dhaka, dashboard	Dhaka city, press	Dhaka, repaired	Barishal, dashboard	Barishal, press	Barishal, repaired	Left unassigned
9 Jul 2023	6,746	596	4,737		4,737	79		675	0
16 Jul 2023	11,231	1,325	6,621		6,805	137	1,278	1,278	0
23 Jul 2023	15,722	8,755	2,772	7,963	9,755	23	1,795	1,795	0
30 Jul 2023	17,561	9,720	2,632	7,675	10,359	0	1,993	1,993	0
6 Aug 2023	18,538	10,002	3,311	7,181	10,958	0	2,355	2,355	0
13 Aug 2023	15,354	8,163	2,572	5,967	8,866	0	1,869	1,869	0
20 Aug 2023	14,324	6,375	2,230	5,818	8,111	1,092	1,586	1,586	0
27 Aug 2023	15,510	7,108	2,421	6,103	8,528	776	1,777	1,777	0
3 Sep 2023	17,641	6,475	2,922	6,473	9,397	2,342	2,342	2,342	0
10 Sep 2023	19,227	6,582	3,736	6,319	10,055	2,578		2,578	263

Dhaka city admissions are added from the week of 23 July. Empty Barishal press cells are weeks where the press-release reading did not pass the checks on every day. In the week of 9 July the leftover of 596 was above 3% and went to Barishal.

After the repair, 791 of the 105,000 admissions could not be assigned to a division, and the largest leftover in a single week is 335. I treat the repaired 2023 values of Dhaka and Barishal as estimates. They are flagged in the data, marked in the figures, and counted in the coverage table of section 3.5.

3.4 Pooled weeks, age labels and the split

Pooled weeks. Outside 2023 a few weeks still differ from the national total by more than 3%. They come in runs of neighbouring weeks where the dashboard put a day or two in the wrong week. Inside each run the single weeks are off, but the run as a whole matches. The daily data carries exact dates, so it is the reference: each division's total over the run is kept and spread over the weeks of the run in proportion to national weekly admissions. This touched 13 weeks.

Age labels. Several age groups were damaged before publication because a spreadsheet read age ranges as dates. "11-15" appears as [Nov-15](#) or as the serial number 42309, and "6-10" as [06-Oct](#). They were mapped back by rule, and any label that is still not one of the 17 standard bands becomes "unknown". This repaired the labels of 30,111 people, almost all of them in 2023. The age charts are a separate count on the dashboard and can differ slightly from the daily totals, so they are used for shares and rates, never as case totals.

Split. Weeks of the epidemiological years 2023 and 2024 are the training set (104 weeks), 2025 is the validation set (52 weeks) and 2026 is the test set (39 weeks). The split is saved in one file that every later notebook and script reads.

3.5 What the cleaned data looks like

Table 7. The four cleaned tables.

Table	Content	Rows	Duplicate keys	Missing cells
national_daily	One row per day, national admissions and deaths	1,370	0	54
national_weekly	One row per week, national	195	0	0
division_weekly	One row per division and week, with weather and population	1,560	0	0
age_sex	Admissions and deaths by year, sex and age group	276	0	0

The 54 missing cells are the 27 unreported days, missing in both admissions and deaths.

Table 8. Coverage by epidemiological year, and how many division-weeks were adjusted in this notebook (query q01).

Year	Weeks	Weeks with all 7 days reported	Days without a report	Division-weeks	Adjusted	Adjusted, %
2023	52	41	19	416	34	8.2
2024	52	48	8	416	9	2.2
2025	52	52	0	416	30	7.2
2026	39	39	0	312	46	14.7

About 8% of all division-weeks (119 of 1,560) carry an adjusted value. In 2026 the share is 14.7%, because several runs of misplaced days fall in that year. The adjustments move admissions between neighbouring weeks or, in 2023, between what the dashboard published and what the press releases show. They do not change any national total.

Table 9. The cleaning log in short. The full log is saved as `data/processed/cleaning_log.csv`.

Rule	Size
Kept only the charts that change with the year filter	4,327 chart rows kept
Removed an invalid date (01-Jan-70) in the 2025 daily chart	6 rows
Days with no report kept as missing, not set to zero	27 days
Removed division values that had no week label	6 values
Matched the dashboard week labels to Sunday weeks by the data	2,016 values
2023: Barishal replaced by press-release values in 8 weeks	10,625 admissions added
2023: Dhaka city corporations added back to Dhaka	91,241 admissions added
2023: Barishal leftover above 3% added in 3 weeks	1,991 admissions added
2023: leftover that could not be assigned	791 admissions (largest week 335)
Runs of neighbouring weeks pooled and split by the national daily data	13 weeks
Dropped the last, unfinished week	1 week
Age labels repaired or set to unknown	30,111 people

4 SQL analysis

Notebook 03_duckdb_sql_analysis.ipynb

The first answers to the public questions come from SQL, run with DuckDB straight on the cleaned Parquet files. DuckDB runs inside the notebook, so there is no database server to set up, and the same queries would run on a SQL warehouse with small changes. There are eleven queries, each saved as a CSV.

Table 10. The eleven queries.

Query	Question	SQL used
q01	How complete is the data in each year?	Two CTEs joined one to one, COUNT(*) FILTER
q02	How big was each year's outbreak?	ARG_MAX, LAG, a scalar CTE for population
q03	At the same point in the year, how does 2026 compare?	Running totals with SUM() OVER
q04	In which months do the cases fall?	SUM(SUM()) OVER, PIVOT
q05	When does the season start, peak and end?	A named WINDOW looking three weeks ahead, filtered aggregates
q06	Which divisions carry the burden?	Share of a partition total, RANK, LAG per division
q07	Do other divisions peak later than Dhaka?	ARG_MAX, self-join of a CTE
q08	Who is admitted and who dies?	Regular expression, CASE banding, ROLLUP
q09	Why work in weeks and not days?	Window average per week, ratio of sums, PIVOT
q10	How many weeks after the weather do cases follow?	Generated list of lags, join on shifted dates, CORR, UNPIVOT
q11	How good is a forecast that needs no model?	Two self-joins on shifted dates, GROUPING SETS

The queries are written one clause per line with aligned aliases. The season query is a fair example. A named window looks three weeks ahead, so that one busy week does not count as the start of a season.

SQL	Notebook 03, query q05: flagging in-season weeks and the first week that starts three in-season weeks in a row (first two CTEs)
WITH params AS (SELECT 1000), flagged AS (SELECT w.year w.week_start w.epi_week w.admitted w.admitted >= p.season_threshold MIN(w.admitted) OVER next_three >= p.season_threshold AND COUNT(*) OVER next_three = 3 w.admitted - LAG(w.admitted) OVER (ORDER BY w.week_start) FROM weeks AS w CROSS JOIN params AS p WINDOW next_three AS (ORDER BY w.week_start ROWS BETWEEN CURRENT ROW AND 2 FOLLOWING)) ...	AS season_threshold AS year, AS week_start, AS epi_week, AS admitted, AS in_season, AS starts_three_week_run, AS weekly_rise

4.1 Burden by year

Table 11. Admissions, deaths and the peak week by epidemiological year (query q02).

Year	Weeks	Admissions	Deaths	Deaths per 100 admissions	Admissions per 100,000	Peak week	Admissions in peak week	Change on previous year, %
2023	52	316,906	1,703	0.54	191.9	17 Sep 2023	20,155	
2024	52	100,971	572	0.57	61.1	3 Nov 2024	7,891	-68.1
2025	52	102,788	417	0.41	62.2	2 Nov 2025	6,835	+1.8
2026 (to 20 Sep)	39	71,949	220	0.31	43.6	20 Sep 2026	11,157	

The 2023 outbreak stands apart: 316,906 admissions and 1,703 deaths. Admissions fell by 68.1% in 2024 and stayed at that level in 2025 (up 1.8%). Deaths per 100 admissions were 0.54 and 0.57 in the first two years and lower after that, 0.41 in 2025 and 0.31 so far in 2026. Over the whole period the series holds 592,614 admissions and 2,912 deaths.

The year 2026 is still running, so it cannot be compared with full years. The fair comparison cuts every year at the same week.

Table 12. Every year cut at week 39, the latest week of 2026 (query q03).

Year	Up to week	Admissions to that week	Deaths to that week	Ratio to 2026	Share of the full year's admissions, %
2023	39	199,239	989	2.77	62.9
2024	39	28,668	166	0.40	28.4
2025	39	45,550	193	0.63	44.3
2026	39	71,949	220	1.00	

By week 39, 2026 had 71,949 admissions. That is 1.58 times the 45,550 of 2025 at the same week and 2.5 times the 28,668 of 2024, but still well below 2023, which had 2.77 times as many. The last column matters for reading 2026: in 2024 only 28.4% of the year's admissions had arrived by week 39, and in 2025 only 44.3%. If 2026 follows either pattern, more than half of its season is still ahead.

4.2 Months and weekdays

Table 13. Share of each year's admissions by month, % (query q04). 2026 is the share of the year so far.

Month	2023	2024	2025	2026
January	0.1	1.1	1.5	2.1
February	0.0	0.4	0.4	0.6
March	0.0	0.3	0.3	0.4
April	0.0	0.4	0.6	1.0
May	0.2	0.7	1.8	0.9
June	1.6	0.7	5.0	3.4
July	11.4	1.9	11.5	14.0
August	25.7	7.3	9.3	24.9
September	23.9	15.6	13.9	52.7
October	19.7	34.3	24.6	
November	14.9	28.0	22.6	
December	2.6	9.3	8.5	

Table 14. Admissions reported per weekday, where 100 is an average day of the same week (query q09).

Weekday	2023	2024	2025	2026
Sunday	109	121	128	112
Monday	107	109	119	114
Tuesday	107	114	110	115
Wednesday	107	110	108	107
Thursday	103	106	99	110
Friday	69	42	51	49
Saturday	97	99	84	94

In 2023 the season was centred on August and September. In 2024 and 2025 it moved to October and November, which together hold 62.3% and 47.2% of the year.

Reporting drops sharply on Fridays, to between 42 and 69 against an average day of 100. A daily model would mostly learn this reporting rhythm. That is the reason the project forecasts weekly totals.

4.3 Divisions

Table 15. Admissions per 100,000 people (rate) and share of national admissions in % (share), by division and year (query q06). 2026 is up to 20 September.

Division	Rate 2023	Rate 2024	Rate 2025	Rate 2026	Share 2023	Share 2024	Share 2025	Share 2026
Dhaka	382.2	131.2	111.0	65.5	53.5	57.5	47.8	40.3
Chattogram	131.2	46.4	45.2	32.0	13.8	15.3	14.6	14.8
Barishal	401.4	95.8	236.9	106.2	11.6	8.6	21.0	13.4
Khulna	192.8	57.2	31.5	67.4	10.6	9.9	5.3	16.3
Rajshahi	92.6	19.2	28.8	21.8	6.0	3.9	5.7	6.2
Mymensingh	62.8	25.1	35.2	32.6	2.4	3.0	4.2	5.5
Rangpur	31.5	8.6	6.0	12.3	1.8	1.5	1.0	3.0
Sylhet	12.6	3.1	3.9	3.0	0.4	0.3	0.4	0.5

Raw counts favour Dhaka because it has the most people, so the rate per 100,000 is the fair comparison. On that measure Barishal had the highest rate in 2023, 2025 and 2026 so far, and Dhaka had the highest in 2024. Dhaka division's share of national admissions was 53.5% in 2023 and 57.5% in 2024, then fell to 47.8% in 2025 and 40.3% in 2026 so far. Its share of the population is 26.8%, so it still carries more than its size would explain. The largest moves are Barishal in 2025, whose share rose by 12.3 points to 21.0%, and Khulna in 2026, up 11.0 points to 16.3%. Sylhet and Rangpur stay low in every year.

4.4 The benchmark every model has to beat

Two forecasts need no model. The **last value** forecast says the coming weeks will look like the latest known week. The **same week last year** forecast repeats the same week one year earlier. Their mean absolute error is the number every model in sections 8 and 9 has to beat. A model that cannot beat "the same as last week" is not worth running, however advanced it is.

Table 16. Mean absolute error of the two no-model forecasts, all divisions together, in admissions per division and week (query q11).

Year	Weeks ahead	Forecasts	Mean weekly admissions	MAE, last value	MAE, same week last year
2025 (validation)	1	416	247.1	46.3	118.9
2025 (validation)	2	416	247.1	76.5	118.9
2025 (validation)	3	416	247.1	105.7	118.9
2025 (validation)	4	416	247.1	134.1	118.9
2026 (test)	1	312	230.6	44.1	141.3
2026 (test)	2	312	230.6	78.4	141.3
2026 (test)	3	312	230.6	114.6	141.3
2026 (test)	4	312	230.6	148.9	141.3

The last value is hard to beat one week ahead (an error of 46.3 in 2025 and 44.1 in 2026) and gets worse quickly with the horizon. The same week last year is poor at every horizon, because the seasons differ so much in size.

5 Exploratory analysis

Notebook 04_exploratory_analysis.ipynb

Notebook 04 looks at every series before any model is fitted. It has two kinds of sections. Those that describe the outbreaks use all four years. Those that decide how to model use 2023 to 2025 only, so that the test year 2026 stays out of every modelling decision.

5.1 Four years of weekly admissions

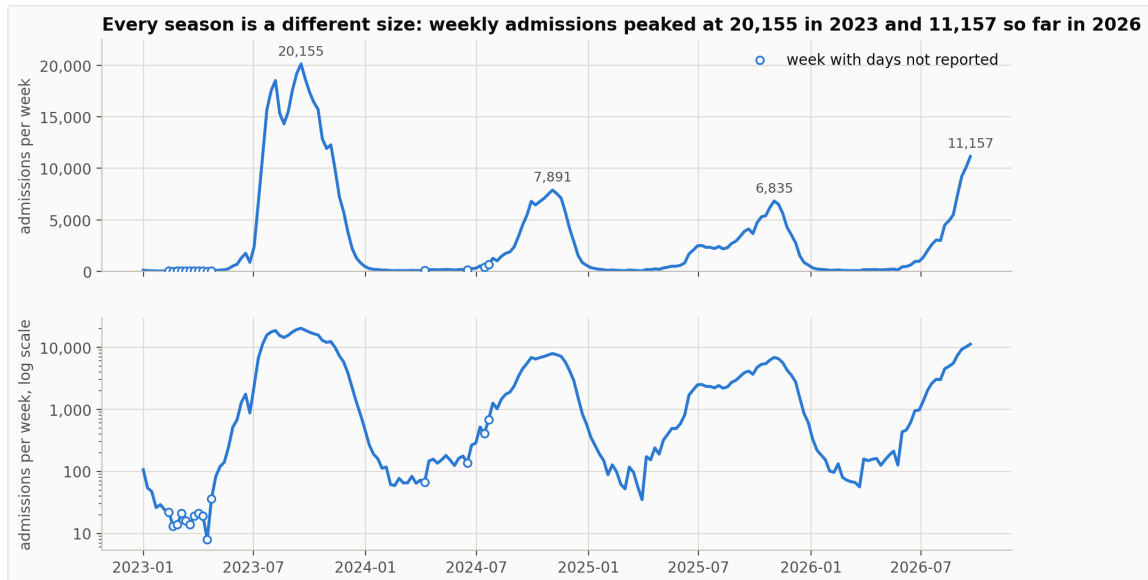


Figure 2. National weekly admissions on a linear scale (top) and a log scale (bottom). Hollow markers are weeks with one or more unreported days.

The four seasons differ a great deal in size. Weekly admissions peaked at 20,155 in the week of 17 September 2023, at 7,891 in the week of 3 November 2024 and at 6,835 in the week of 2 November 2025. In 2026 the highest week so far is the latest one, the week of 20 September, with 11,157. On the log scale the series moves in proportions: it rises and falls by similar factors in every year. That is why the later models work on log counts.

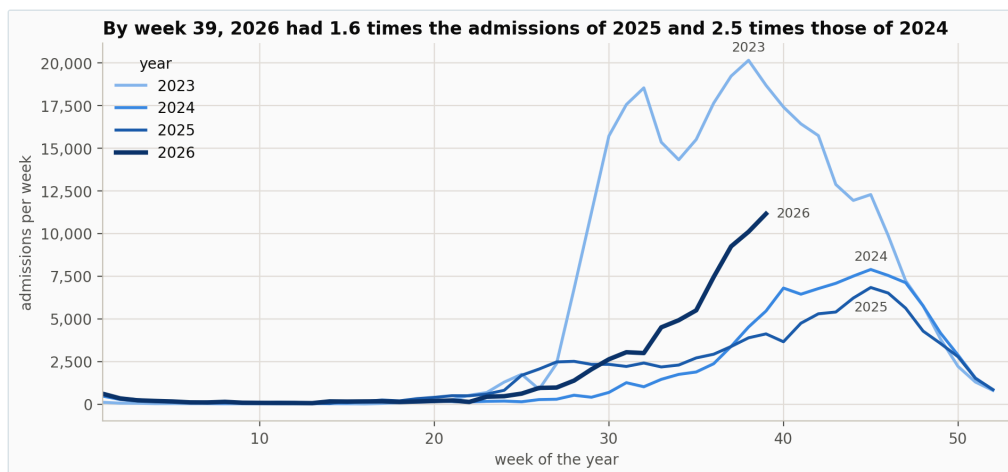


Figure 3. Each year against the week of the year. 2026 stops at the last complete week.

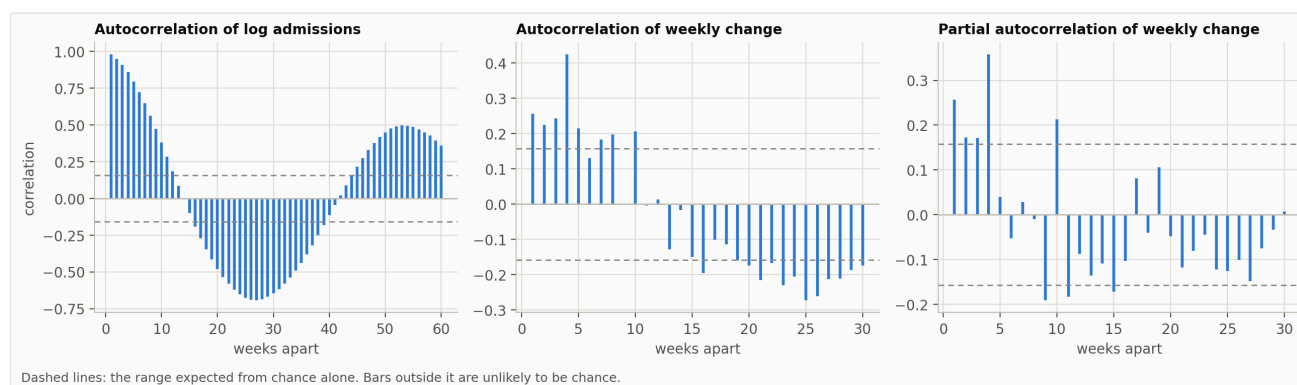
5.2 Stability and memory of the series

Classical time series models need a series whose level does not drift. Two tests with opposite starting assumptions were run on the national log series of 2023 to 2025. The ADF test assumes the series drifts, and the KPSS test assumes it is stable. Both agree that the level is stable from year to year. The yearly cycle is still in the series and is handled separately, by seasonal terms.

Table 17. Stationarity tests on national log admissions, 2023 to 2025.

Series	Weeks	ADF p-value	KPSS p-value	Reading
log admissions	156	0.000	0.10 or above	stable
change from last week	155	0.002	0.10 or above	stable
change from same week last year	104	0.036	0.10 or above	stable

A small ADF p-value points to a stable level. A small KPSS p-value points to drift. The KPSS p-value is reported by the software as 0.1 when it lies at or above that bound.

**Figure 4.** Autocorrelation of national log admissions (left), and autocorrelation and partial autocorrelation of the week-to-week change (middle, right). 2023 to 2025.

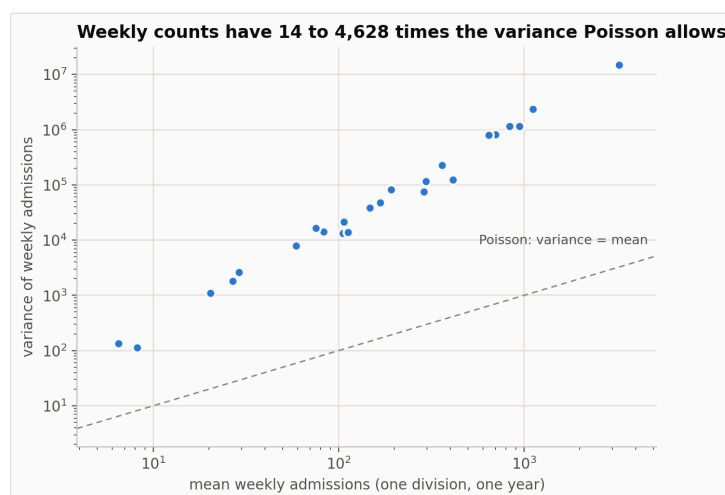
The left panel is dominated by the yearly cycle: the bars fall slowly, turn negative about half a year apart and come back. After taking week-to-week changes only a few short lags stand outside the band. The table below asks the same question in a simpler way, per division.

Table 18. Correlation of each division's log admissions with its own past, 2023 to 2025.

Division	1 week ago	2 weeks ago	3 weeks ago	4 weeks ago	8 weeks ago	52 weeks ago
Barishal	0.98	0.95	0.92	0.88	0.66	0.75
Dhaka	0.98	0.95	0.92	0.88	0.60	0.85
Khulna	0.97	0.94	0.90	0.85	0.59	0.89
Chattogram	0.96	0.94	0.91	0.88	0.64	0.84
Rajshahi	0.97	0.94	0.90	0.85	0.56	0.70
Mymensingh	0.96	0.93	0.89	0.83	0.55	0.83
Rangpur	0.94	0.89	0.85	0.79	0.49	0.69
Sylhet	0.92	0.88	0.83	0.77	0.47	0.35

The last one to four weeks carry most of the information (a median correlation of 0.96 one week back and 0.85 four weeks back). The same week a year earlier carries less (0.79), and much less in Sylhet (0.35), where there is hardly a regular season.

5.3 Counts vary far more than a Poisson model allows

**Figure 5.** Variance against mean of weekly admissions, one point per division and year (log scales). The dashed line is where a Poisson model would lie.

A Poisson model assumes the variance of a count equals its mean. Here the variance is 14 to 4,628 times the mean, depending on the division and year. So the count models in this project use a negative binomial distribution, and the others work on the log scale.

Table 19. Weekly admissions by division, 2023 to 2025.

Division	Mean weekly admissions	Weeks with zero admissions	Variance over mean
Barishal	428.3	13	875
Dhaka	1,769.9	0	4,081
Khulna	314.3	24	1,112
Chattogram	474.2	5	1,073
Rajshahi	183.4	35	554
Mymensingh	96.5	20	220
Rangpur	52.0	56	188
Sylhet	13.8	65	55

5.4 Divisions in time

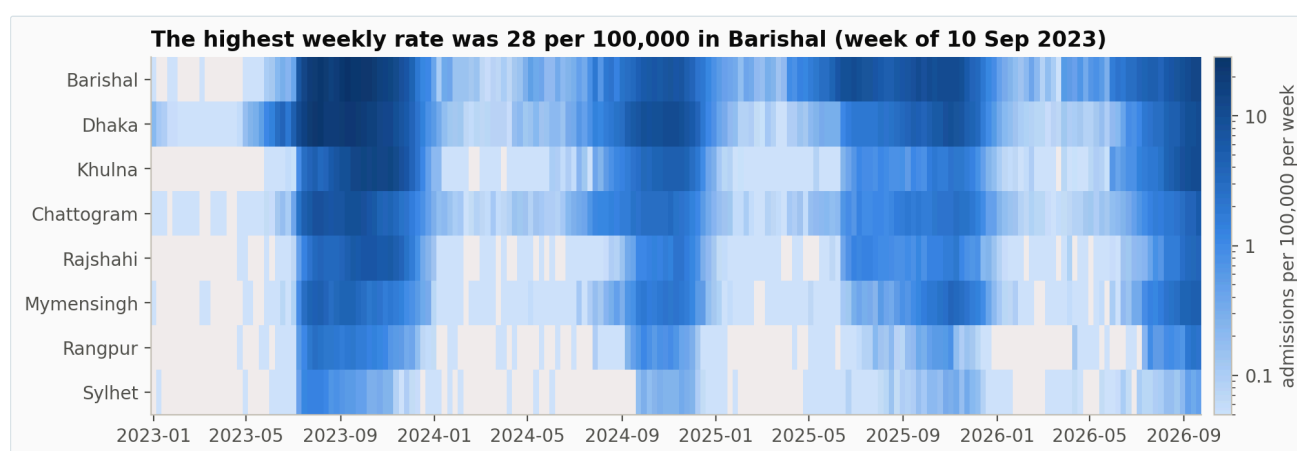


Figure 6. Weekly admissions per 100,000 people by division (log colour scale). Rows are ordered from the highest overall rate to the lowest.

The highest weekly rate in the data is 28.3 admissions per 100,000 people, in Barishal in the week of 10 September 2023. The heatmap also shows what the yearly table could not: Barishal's 2025 season started months before the others, and Sylhet and Rangpur have long stretches with almost no admissions.

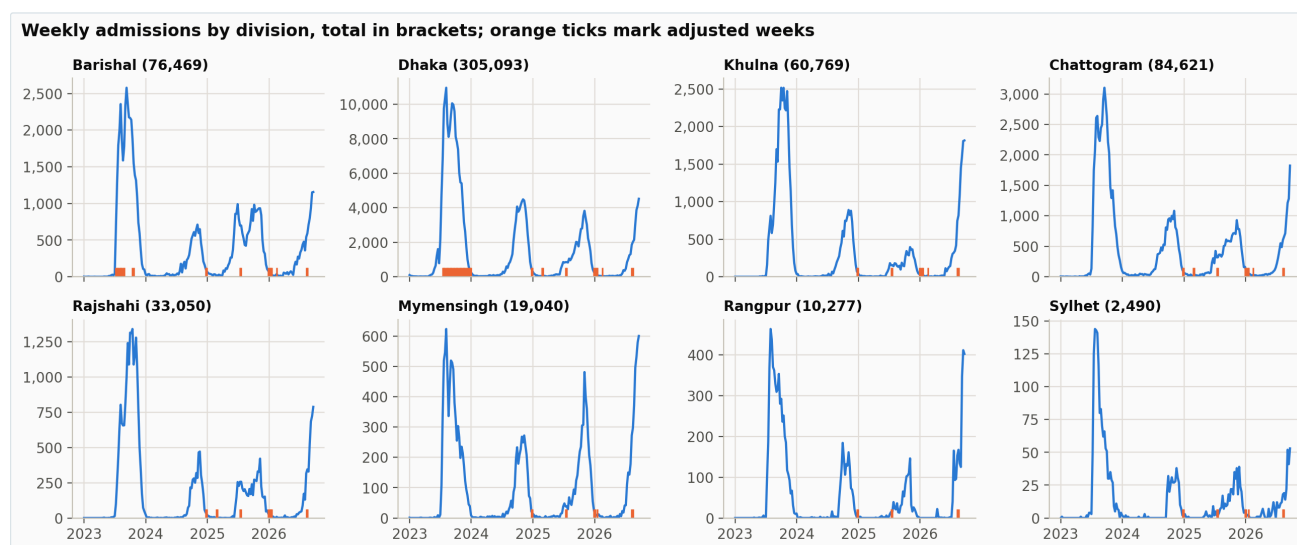


Figure 7. Weekly admissions by division, each panel on its own scale. Orange ticks mark weeks whose values were adjusted in notebook 02.

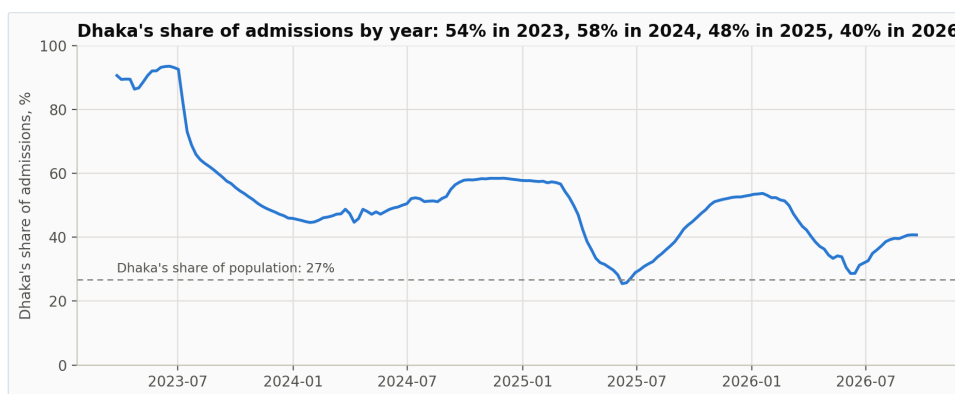


Figure 8. Dhaka division's share of national admissions over a moving window of 13 weeks. The dashed line is its share of the population.

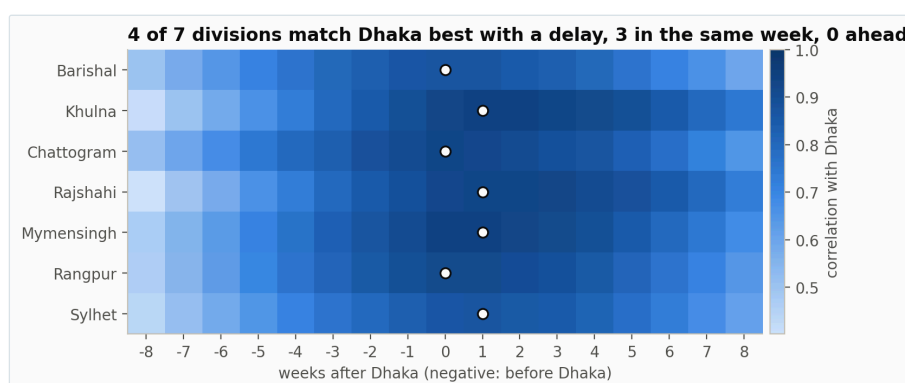


Figure 9. Correlation of each division's log admissions with Dhaka's, shifted from 8 weeks earlier to 8 weeks later. The dot marks the best match. 2023 to 2025.

If the wave reached other divisions some weeks after Dhaka, Dhaka's latest count would be an early warning for them. The data gives this idea little support. Three divisions match Dhaka best in the same week and four with a delay of one week, and the gain from shifting is tiny (for Khulna the correlation is 0.940 at one week against 0.928 in the same week). All divisions share one season, and none runs clearly behind Dhaka. Dhaka's recent weeks were still tested as model features in section 8, and they did not help.

5.5 Who is admitted and who dies

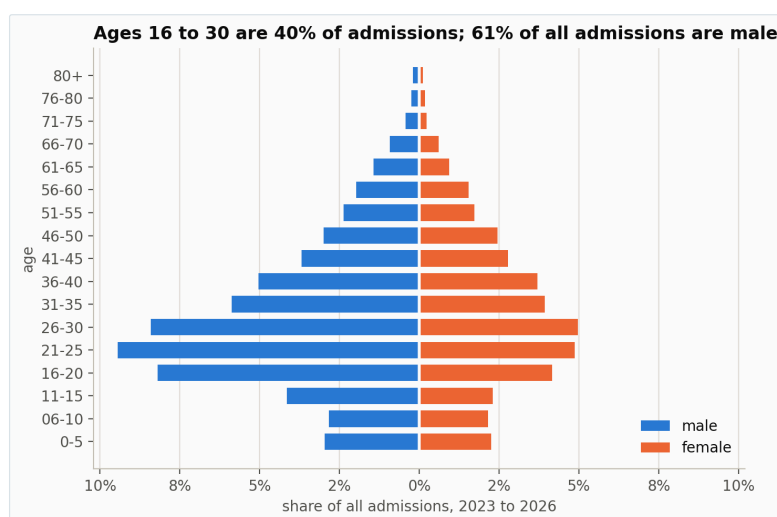


Figure 10. Share of all admissions, 2023 to 2026, by age group and sex.

Young adults carry the largest share. People aged 16 to 30 are 40% of admissions over the four years, and 61% of all admissions are male.

Deaths look different. Women are 37% to 40% of admissions in each year but 48% to 57% of deaths. And deaths per 100 admissions rise steeply with age: in 2023 from 0.31 below age 16 to 1.75 above age 60, and in 2024 from 0.39 to 2.11. These are rates among hospital cases. They can reflect who gets admitted and how late, as well as the risk of the disease itself, so I report the pattern and do not explain it.

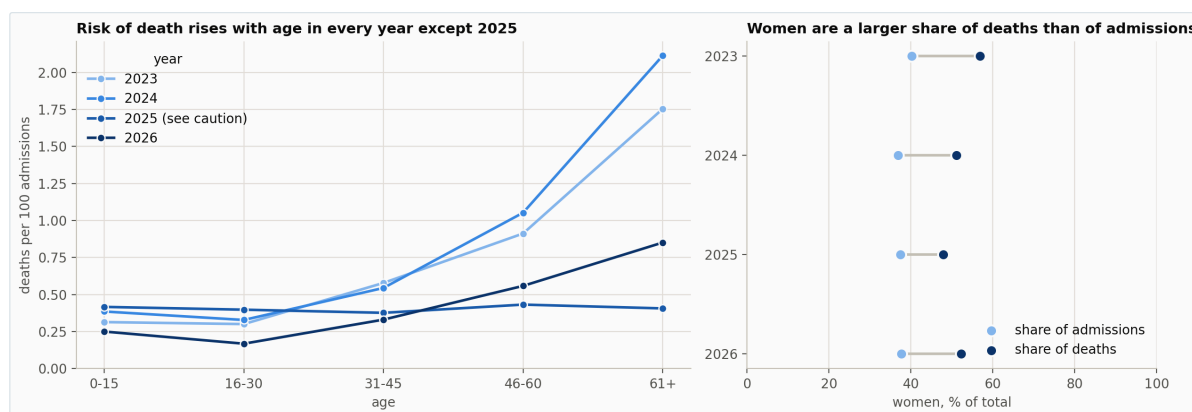


Figure 11. Deaths per 100 admissions by age band and year (left), and women's share of admissions against their share of deaths (right).

Table 20. Share of admissions and of deaths over the four years, and deaths per 100 admissions by year (query q08).

Age band	Share of admissions, %	Share of deaths, %	2023	2024	2025	2026
0-15	16.6	11.4	0.31	0.39	0.42	0.25
16-30	40.0	24.9	0.30	0.33	0.40	0.17
31-45	25.0	25.8	0.58	0.54	0.38	0.33
46-60	13.1	21.7	0.91	1.05	0.43	0.56
61+	5.3	16.1	1.75	2.11	0.41	0.85
all ages	100.0	100.0	0.54	0.57	0.40	0.30

The age charts are a separate count on the dashboard, so they are used for shares and rates, not totals. People with no usable age (12 admissions, 3 deaths) are left out of the bands.

A caution on 2025

In 2023, 2024 and 2026 the death rate rises with age. In 2025 the dashboard shows almost the same rate in every band (0.38 to 0.43), and no death is recorded above age 75 although more than 700 people of that age were admitted. The total of 413 deaths matches the daily series, so this is how the source records the ages of the 2025 deaths, not a processing error. The DGHS press release of 31 December 2025 shows the same table, but it comes from the same system and does not confirm it independently. I do not use the 2025 age pattern of deaths for any conclusion.

5.6 Weather and the shared season

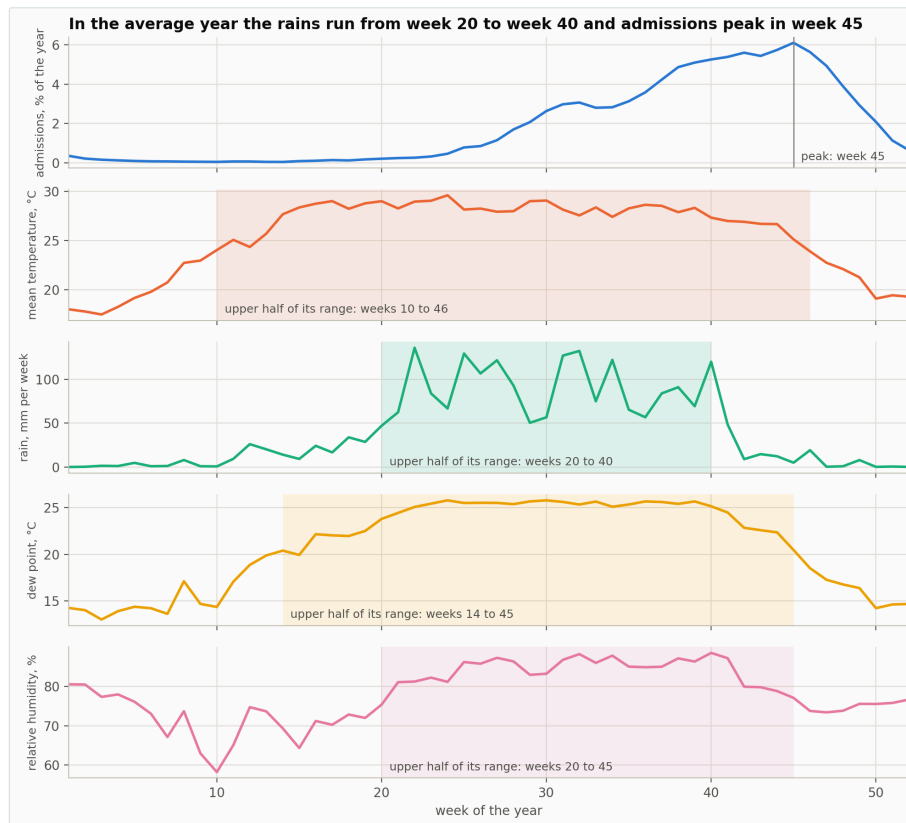


Figure 12. The average year, 2023 to 2025: admissions as a share of the year, and each weather variable, by week of the year. Shaded bands mark the weeks in the upper half of each variable's yearly range.

In the average year the rains are in the upper half of their range from week 20 to week 40 and admissions peak in week 45. Read naively, cases follow the weather by two months or more. The raw correlations say the same: across the eight divisions, log admissions correlate at 0.5 to 0.8 with dew point, temperature and rainfall six to ten weeks earlier.

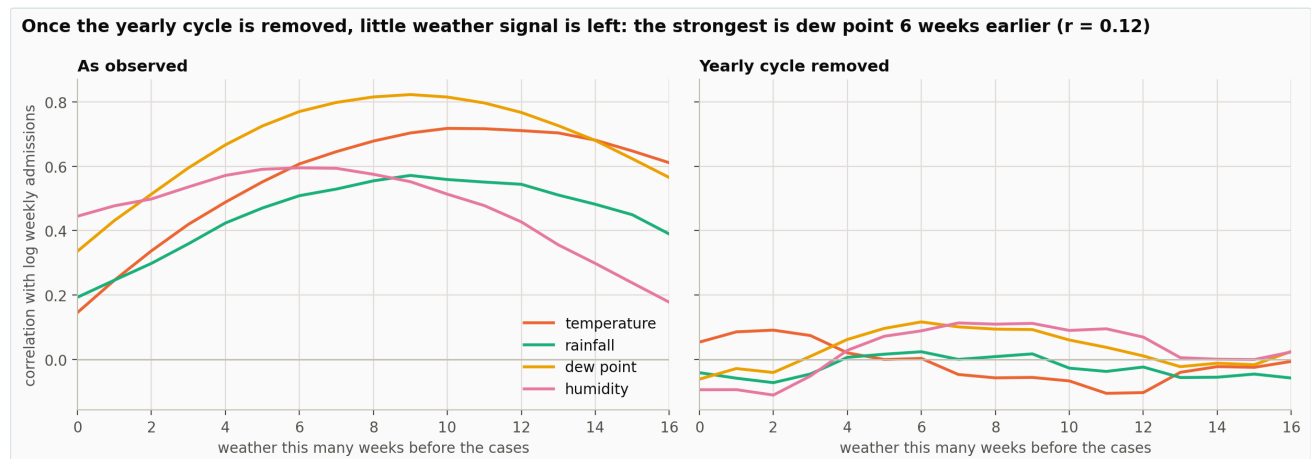


Figure 13. Correlation between log weekly admissions and weather 0 to 16 weeks earlier, mean of the eight divisions: as observed (left) and after the yearly cycle is removed from both series (right). 2023 to 2025.

But dengue and the monsoon share a yearly cycle, so they correlate whether or not weather drives anything. The right panel removes a smooth annual curve from both series and asks the sharper question: when a week is wetter or hotter than usual for the time of year, are admissions some weeks later higher than usual? Almost nothing survives. The strongest remaining correlation is 0.12, for dew point six weeks earlier, and in some divisions it is zero or negative. Most of what weather seems to explain is simply the calendar. Section 7 estimates the effect properly, and section 8 tests whether weather improves a forecast.

5.7 What this means for the models

- **Scale.** The series moves in proportions and its variance is far above its mean, so models work on log counts or use a negative binomial distribution.
- **Season.** One yearly cycle dominates. With three seasons to learn from, the season is described by a few smooth terms (Fourier terms, week of the year), not by a full 52-week seasonal model.
- **Memory.** The last one to four weeks carry most of the information. Lag features start from these.
- **Geography.** Divisions differ a hundredfold in size and do not run behind Dhaka, so the machine learning models are fitted on all divisions together with the division as a feature.
- **Weather.** It is tested as an optional extra and kept only if it improves the 2025 forecasts.
- **Unequal seasons.** 2023 was about three times the size of 2024 and 2025. Errors are reported per year, so one large season cannot hide a poor fit elsewhere.

6 Seasonality

R script 06_seasonality.R

This script asks when the dengue season starts, peaks and ends, and whether that differs by year and by division. All four years are described. There are only three complete seasons, so changes between years are described and not tested: no trend test is run.

6.1 How strong is the yearly cycle?

Each series is split on the log scale into a trend, one fixed yearly shape and a remainder, using an STL decomposition with a periodic seasonal window and a robust fit [9]. Seasonal strength is one minus the variance of the remainder over the variance of season plus remainder [10]. It runs from 0 (no yearly cycle) to 1 (the series is all cycle).

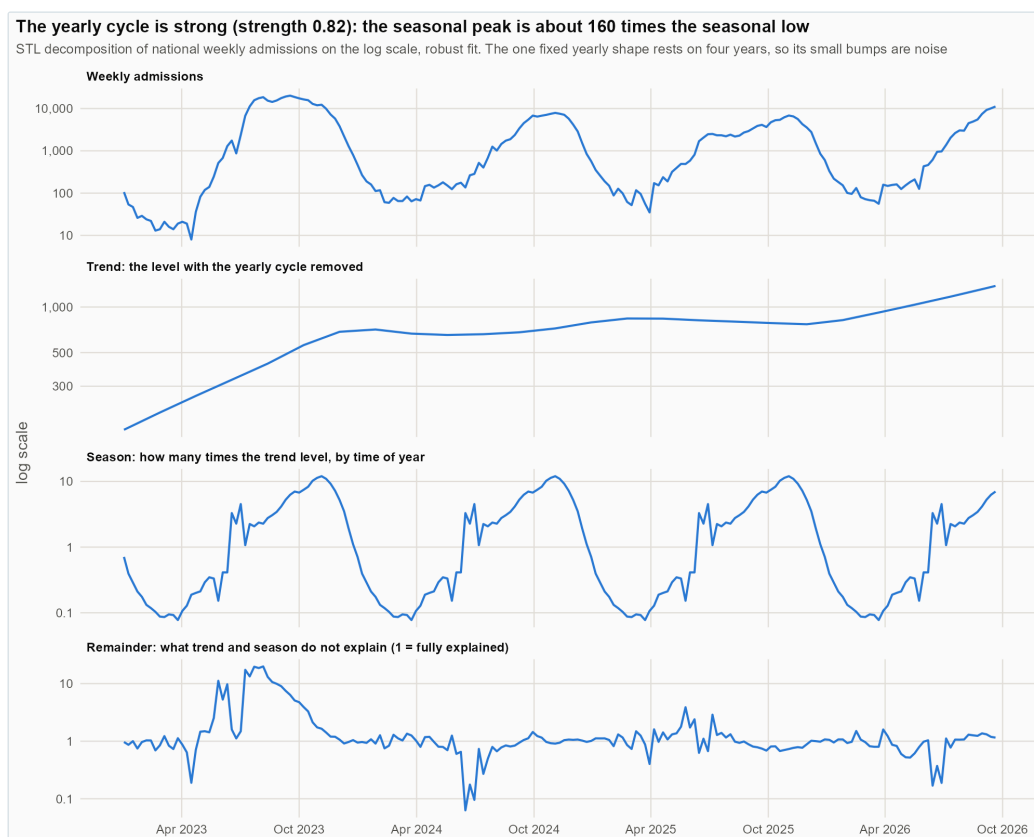


Figure 14. STL decomposition of national weekly admissions on the log scale: the series, the trend, the yearly shape and the remainder.

Table 21. Strength of the yearly cycle and of the trend by area.

Area	Seasonal strength	Trend strength	Seasonal peak over low	Peak week	Low week
Bangladesh	0.82	0.05	160	45	13
Dhaka	0.83	0.00	220	40	13
Chattogram	0.81	0.37	85	45	10
Barishal	0.71	0.36	94	47	13
Khulna	0.84	0.30	280	47	14
Rajshahi	0.83	0.23	520	46	19
Mymensingh	0.80	0.09	470	45	9
Rangpur	0.80	0.10	270	40	23
Sylhet	0.58	0.08	45	40	10

For Bangladesh the seasonal strength is 0.82 and the trend strength is 0.05. The seasonal peak falls in week 45 and the low in week 13, and the peak is about 160 times the low. Seven divisions have a strength of 0.71 to 0.84. Sylhet is the exception at 0.58, because it has too few cases to form a regular season. The remainder panel shows the limit of this description: one fixed yearly shape cannot follow the 2023 outbreak, which sits far above it for months.

6.2 Onset, peak and length by a fixed rate

A week is in season when national admissions reach 1,000, which is 0.61 per 100,000 people. Each division uses the same rate per person, so the threshold is 268 admissions a week in Dhaka and 55 in Barishal. The onset is the first week that starts three in-season weeks in a row, the same rule as SQL query q05.

Table 22. The national season by year, at a threshold of 1,000 admissions a week.

Year	Onset	Onset week	Peak	Peak week	Admissions at peak	Onset to peak, weeks	Weeks in season	Share of year's admissions in season, %
2023	2 Jul 2023	27	17 Sep 2023	38	20,155	11	27	98.8
2024	28 Jul 2024	31	3 Nov 2024	45	7,891	14	21	93.6
2025	15 Jun 2025	25	2 Nov 2025	45	6,835	20	27	93.3
2026 (running)	5 Jul 2026	28	20 Sep 2026	39	11,157	11	12	90.3

2026 is still running: its peak is the highest week so far.

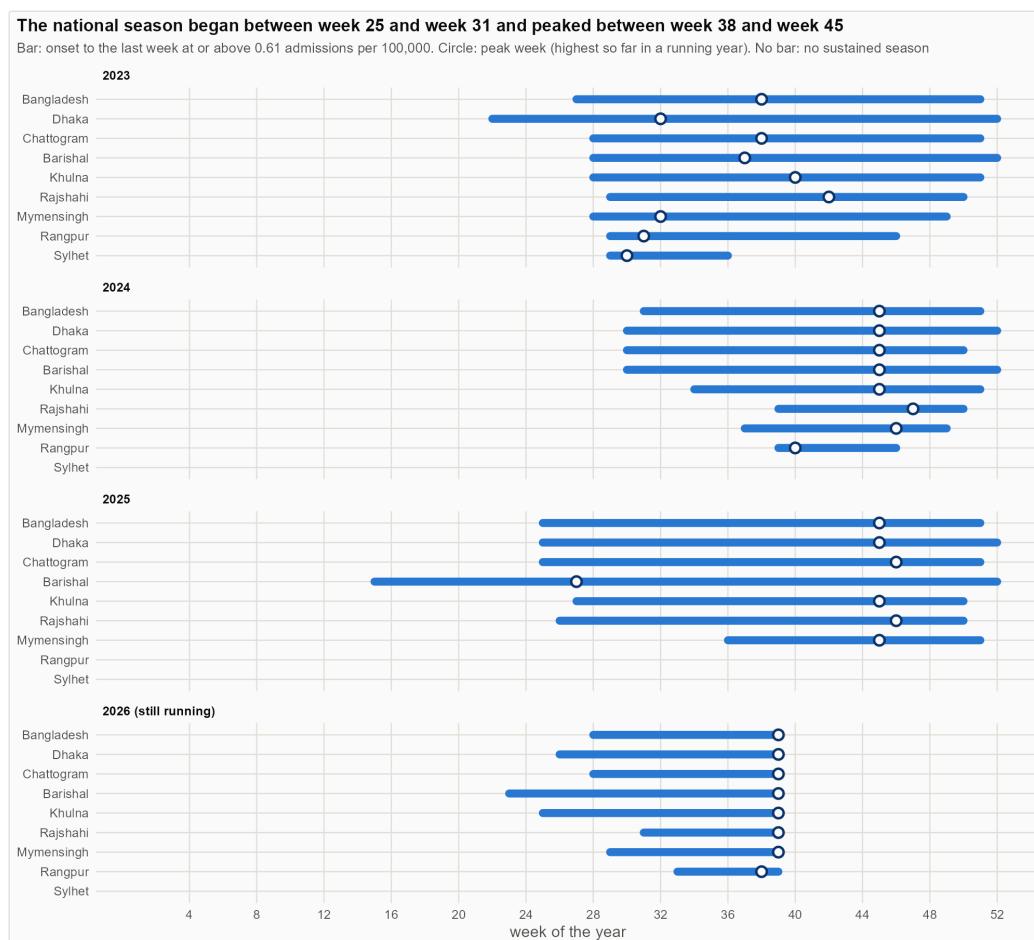


Figure 15. Season from onset to the last in-season week, by area and year. The circle is the peak week. No bar means no sustained season at that rate.

The national season began on 2 July in 2023, 28 July in 2024, 15 June in 2025 and 5 July in 2026. It lasted 27 weeks in 2023 and 2025 and 21 weeks in 2024, and it has run for 12 weeks so far in 2026. More than 93% of each complete year's admissions fall inside the season. The time from onset to peak was 11 weeks in 2023 but 20 weeks in 2025, a long, slow season.

The divisions do not always move together. Barishal's 2025 season began in the week of 6 April, ten weeks before Dhaka's, peaked at the end of June and lasted 39 weeks. Rangpur had no sustained season in 2025, and Sylhet had none in 2024, 2025 or 2026.

6.3 Timing without a threshold

A threshold favours large seasons, so a second measure uses none. Half of the year's admissions had arrived by week 37 in 2023, week 43 in 2024 and week 41 in 2025. The middle 80% of the season took 15, 14 and 21 weeks.

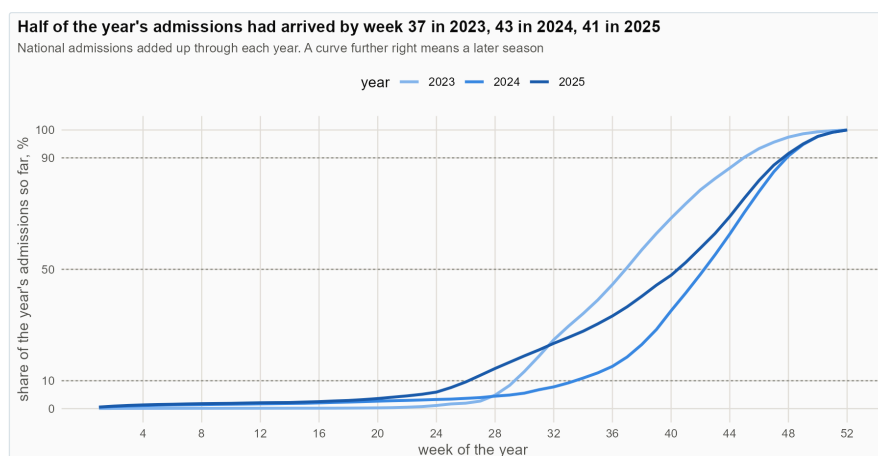


Figure 16. National admissions added up through each complete year, as a share of the year's total.

Table 23. Weeks by which 10%, 50% and 90% of the year's national admissions had arrived.

Year	10% by week	50% by week	90% by week	Weeks from 10% to 90%
2023	30	37	45	15
2024	34	43	48	14
2025	27	41	48	21

Table 24. Middle of each division's season in weeks after Dhaka's (negative means before Dhaka).

Division	2023	2024	2025
Chattogram	+1	-2	-2
Barishal	+1	0	-5
Khulna	+5	+1	0
Rajshahi	+4	+2	-3
Mymensingh	0	+1	+2
Rangpur	0	0	0
Sylhet	-3	+2	+2

On this measure the divisions do not run behind Dhaka in any steady way. Over all divisions and complete years the median is 0 weeks and the range is 5 weeks before to 5 weeks after. Khulna and Rajshahi were 5 and 4 weeks later than Dhaka in 2023, but not by nearly as much in the two years after.

6.4 How much does the onset depend on the threshold?

Table 25. National onset week at half the threshold, the threshold and double the threshold.

Threshold, admissions a week	2023	2024	2025	2026
500	22	30	23	25
1,000	27	31	25	28
2,000	27	36	26	29

2024 had the latest onset at every threshold. Which year was earliest depends on the threshold: 2023 at 500 admissions a week, 2025 at 1,000 and 2,000. So the only firm statement about onset is that the 2024 season started later than the others. I would not read a shift towards earlier or later seasons into three years.

7 Weather and dengue

R script 07_weather_dlnm.R

Section 5 showed that raw correlations between weather and admissions mostly reflect the shared season. This script estimates what is left once the season is accounted for, and how many weeks the effect takes to arrive. It explains; it does not forecast, and no forecasting choice is taken from it. For that reason all four years are used.

7.1 Model

The method is a distributed lag non-linear model (DLNM) [11, 12]: a negative binomial regression of weekly admissions in the eight divisions, with the log of population as an offset. Temperature and rainfall each enter through a cross-basis, which is a curve over the variable's own values combined with a curve over the lag, from 0 to 12 weeks. Both curves are natural splines. The knots for temperature are at the thirds of its distribution, and for rainfall at the median and the 90th percentile, because many weeks have no rain at all. Humidity moves with rainfall, since both mark the wet season, so it is tested in place of rainfall and not alongside it.

Three models are fitted, each adding one layer of control.

- **M1, weather only.** Division differences and population, nothing else. Weather and the season are mixed together.
- **M2, plus season and year.** Each division gets its own yearly cycle (two waves) and its own level per year. Weather is compared with what is usual for that time of year.
- **M3, plus last week's admissions.** This also allows for the outbreak's own momentum, since dengue cases breed more cases. It is the main model.

R Script 07: the cross-basis of each weather variable and the three sets of controls (line breaks added)

```
basis <- function(v, maxlag = MAXLAG) crossbasis(panel[[v]], lag = maxlag, group = panel$division,
  argvar = list(fun = "ns", knots = quantile(used[[v]], if (v == "rainfall") c(0.50, 0.90) else c(1, 2) / 3)),
  arglag = list(fun = "ns", knots = logknots(maxlag, 2)))

SEASON <- "division * factor(epi_year) + division:(s1 + c1 + s2 + c2)"
CONTROLS <- c("M1 weather only" = "division", "M2 + season and year" = SEASON,
  "M3 + last week's admissions" = paste(SEASON, "+ log_last_week"))
# y ~ cb_temperature + cb_rainfall + <control> + offset(log(population)), fitted with MASS::glm.nb
```

Table 26. Fit of the three weather models.

Model	Division-weeks	Parameters	AIC	Theta	Residual autocorrelation at lag 1
M1 weather only	1,552	32	15,206	0.98	0.79
M2 + season and year	1,552	88	14,696	1.61	0.69
M3 + last week's admissions	1,552	89	13,617	4.64	0.05

The last column decides which intervals can be trusted. In M1 and M2 the residuals of neighbouring weeks are strongly related (0.79 and 0.69), so their confidence intervals are too narrow. In M3 the residual autocorrelation is 0.05, and its intervals can be read as they stand. The price is that last week's admissions already carry the effect of earlier weather, so M3 gives a conservative estimate of what weather adds.

7.2 Cool, hot, dry and wet spells against typical weeks

The headline contrast compares 13 weeks in a row at a low level (the 10th percentile) or a high level (the 90th percentile) with 13 typical weeks (the median). For temperature the three levels are 19.1, 27.1 and 29.5 °C. For rainfall they are 0, 20.4 and 124.9 mm a week.

Table 27. Cumulative relative risk of admissions over lags 0 to 12 weeks, with 95% confidence intervals.

Contrast (level against typical)	M1 weather only	M2 + season and year	M3 + last week's admissions	Clear in M3
Cool spell: temperature 19.1 against 27.1 °C	0.050 (0.028 to 0.090)	0.017 (0.004 to 0.069)	0.23 (0.093 to 0.57)	yes
Hot spell: temperature 29.5 against 27.1 °C	1.26 (0.99 to 1.60)	1.46 (0.92 to 2.32)	1.14 (0.85 to 1.54)	no
Dry spell: rainfall 0.0 against 20.4 mm a week	0.39 (0.29 to 0.53)	0.42 (0.24 to 0.70)	0.64 (0.46 to 0.90)	yes
Wet spell: rainfall 124.9 against 20.4 mm a week	9.46 (5.62 to 15.90)	2.57 (1.05 to 6.30)	1.17 (0.66 to 2.07)	no

A relative risk of 1 means no difference from typical weather. "Clear" means the 95% interval does not include 1.

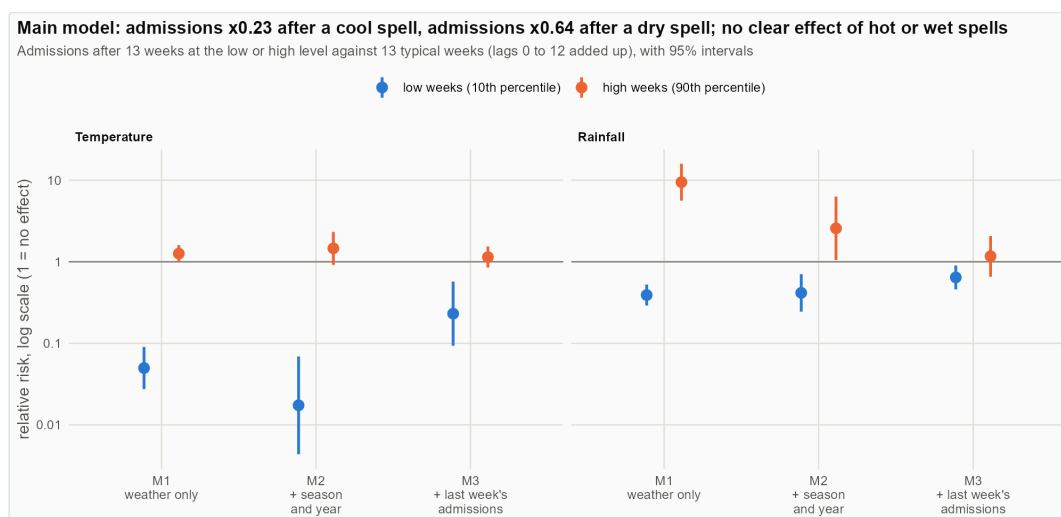


Figure 17. The same contrasts across the three models, on a log scale.

The table shows how much of the apparent weather effect was the season. In M1 a wet spell seems to multiply admissions by 9.46. With the season and year in the model that falls to 2.57, and with last week's admissions it is 1.17 with an interval from 0.66 to 2.07, which is no clear effect.

Two results stay clear in the main model. After a cool spell admissions are 0.23 times those after typical weeks (95% CI 0.093 to 0.57), and after a dry spell 0.64 times (0.46 to 0.90). Hot spells (1.14, 0.85 to 1.54) and wet spells show no clear effect. A caution belongs here: a 13-week cool or dry spell only happens in winter, so these contrasts lean on the model's season terms to separate weather from the time of year.

7.3 When does the effect arrive?

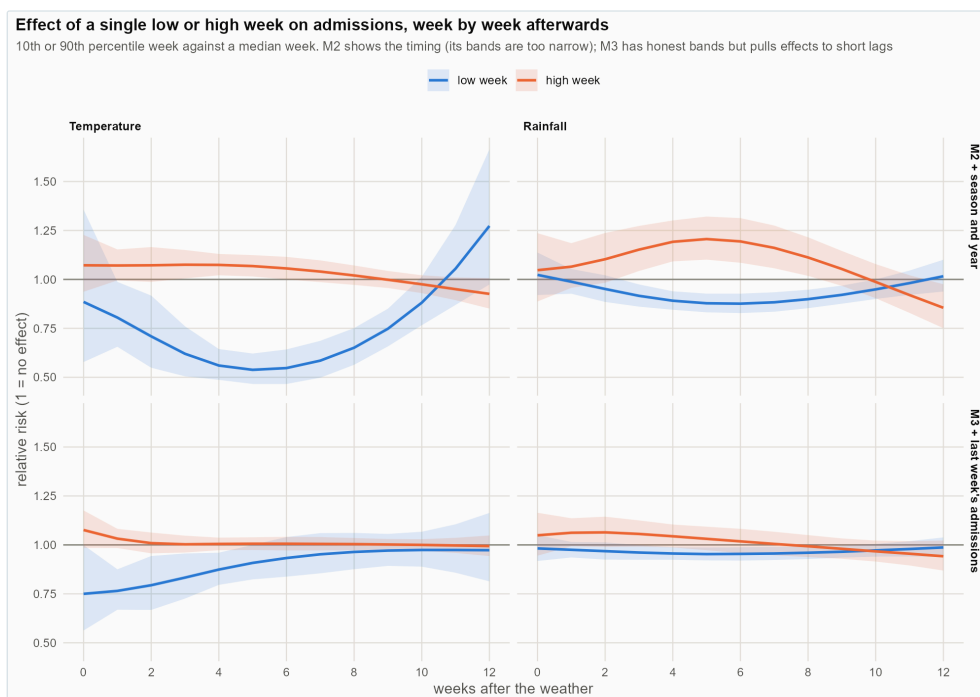


Figure 18. Relative risk in each of the following 12 weeks after one low week or one high week, against a median week. Top: M2. Bottom: M3.

M2 shows the timing, although its bands are too narrow. After one cool week the largest reduction comes 5 weeks later (relative risk 0.54, 0.47 to 0.62). After one wet week the largest increase also comes at 5 weeks (1.21, 1.10 to 1.32), and after one dry week the largest reduction at 6 weeks (0.88, 0.83 to 0.93). A delay of about five to six weeks is what the biology would suggest: time for mosquitoes to breed, for the virus to develop inside the mosquito, and for an infected person to fall ill and reach a hospital. In M3 the effects move to the shortest lags and shrink, because last week's admissions have already absorbed what the earlier weather did.

7.4 Shape of the relationship

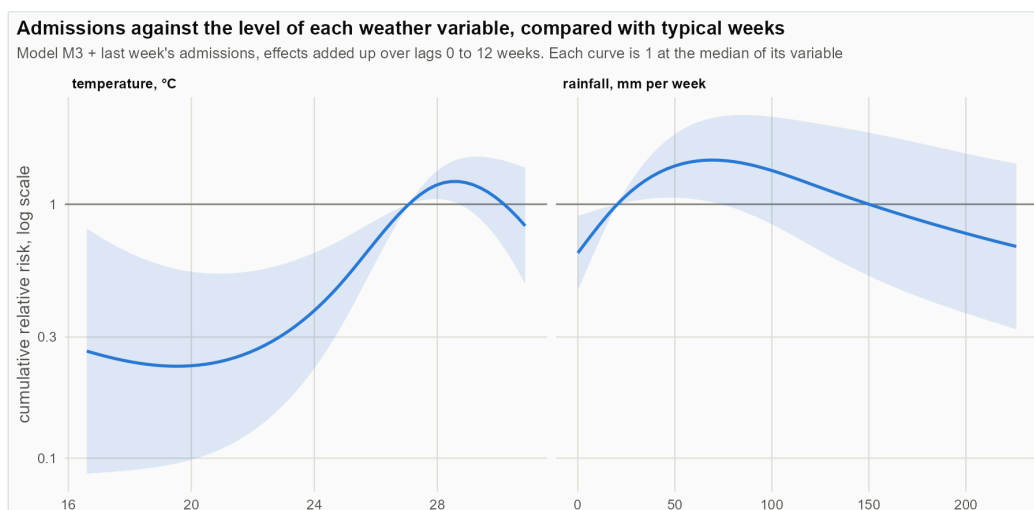


Figure 19. Cumulative relative risk over lags 0 to 12 weeks across the range of temperature and of rainfall, main model, against the median of each variable.

For temperature the curve is low at cool values and rises to a peak near 28 °C, where the relative risk is 1.22 (1.05 to 1.41). Above that it flattens and the interval includes 1. This shape agrees with laboratory-based work, which puts the best temperature for dengue transmission by *Aedes aegypti* near 29 °C [7]. For rainfall the risk is raised at moderate amounts (1.38 at 46 mm a week, 1.06 to 1.79, and 1.49 at 69 mm, 1.02 to 2.18) and uncertain at heavier rainfall, where the interval is wide and includes 1.

7.5 Do the results hold under other reasonable choices?

Table 28. Cumulative relative risks of the main model under six other choices.

Model choice	AIC	Cool spell	Hot spell	Dry spell	Wet spell	Low humidity	High humidity
main model as fitted	13,617	0.23 (0.093 to 0.57)	1.14 (0.85 to 1.54)	0.64 (0.46 to 0.90)	1.17 (0.66 to 2.07)	not in model	not in model
lags 0 to 8 weeks	13,629	0.38 (0.19 to 0.74)	1.00 (0.80 to 1.26)	0.75 (0.58 to 0.96)	1.29 (0.86 to 1.94)	not in model	not in model
lags 0 to 16 weeks	13,591	0.12 (0.039 to 0.38)	1.18 (0.81 to 1.72)	0.70 (0.46 to 1.07)	0.70 (0.33 to 1.47)	not in model	not in model
simpler season (one wave)	13,629	0.38 (0.24 to 0.61)	1.03 (0.80 to 1.32)	0.64 (0.48 to 0.85)	1.20 (0.70 to 2.06)	not in model	not in model
finer season (three waves)	13,608	0.22 (0.086 to 0.56)	0.97 (0.71 to 1.33)	0.75 (0.53 to 1.07)	0.85 (0.47 to 1.56)	not in model	not in model
humidity in place of rainfall	13,571	0.49 (0.16 to 1.47)	0.75 (0.51 to 1.09)	not in model	not in model	1.23 (0.84 to 1.82)	0.41 (0.26 to 0.63)
all three weather variables	13,567	0.44 (0.15 to 1.34)	0.83 (0.56 to 1.22)	0.50 (0.34 to 0.73)	2.26 (1.14 to 4.47)	1.68 (1.10 to 2.56)	0.28 (0.16 to 0.49)

The cool-spell result keeps its direction under all seven choices, from 0.12 to 0.49, and is clear in five of them. The dry-spell result also keeps its direction (0.50 to 0.75) and is clear in four of the six models that include rainfall. The hot-spell and wet-spell results change direction between choices, so they are not firm.

How I read this section

Cooler and drier weather than usual is followed by fewer dengue admissions over the next weeks, with the largest single-week effect about five to six weeks later. There is no firm evidence here that hot or very wet weeks add admissions beyond what the season already predicts. These are associations in surveillance data, measured with one weather point per division. They are not causal estimates, and they say nothing about individual risk.

8 Machine learning forecasts

Notebook 05_ml_forecasts.ipynb

This notebook forecasts weekly admissions in each division one to four weeks ahead with two tree models, XGBoost [13] and a random forest [14], and checks them against the two no-model forecasts of section 4.4.

8.1 What the model sees

The data is reshaped so that one row is one forecast: one division, one origin week in which the forecast is made, and one horizon. Everything on the row is known in the origin week. The table has 6,096 rows and 23 candidate features. The target is admissions in the target week on the log scale, and forecasts are turned back into admissions before any error is measured.

Table 29. Feature groups.

Group	Features	Used
Recent weeks	Admissions this week and 1, 2 and 3 weeks ago (log scale)	Always
Growth	Change over the last week and over the last three weeks	Always
Last year	Admissions in the target week one year earlier, and how much they changed over the same weeks	Always
Season	Week of the year of the target week, as two smooth waves	Always
Division	Which division	Always
Dhaka	Dhaka's latest week and growth	Tested
Weather	Rain, temperature, dew point and humidity: mean of the last 4 weeks and of the 4 weeks before	Tested

The most common way to get a flattering forecast is to let a feature see the future. The notebook tests this in code. It rebuilds the feature table from data that stops at one week in mid 2025 and checks that the features for that week are identical to those in the full table. If any feature looked ahead, the two would differ and the notebook would stop.

Python Notebook 05: the leakage test (line breaks added)

```
F, LAST, CUT = build(weekly), weekly["week_start"].max(), pd.Timestamp("2025-06-29")
at_cut = [t[t["origin_week"] == CUT].set_index(["division", "horizon_weeks"])[SETS["+ Dhaka and weather"]]
           for t in (F, build(weekly[weekly["week_start"] <= CUT]))]
assert len(at_cut[0]) == 32 and at_cut[0].equals(at_cut[1]), "a feature uses information from after the origin week"
# printed: 6,096 rows, 23 candidate features; leakage test passed for the week of 29 Jun 2025
```

8.2 Choosing features and settings on 2025

Each candidate is fitted on target weeks in 2023 and 2024 and scored on target weeks in 2025, with one model per horizon. Four things are varied.

- **Model.** XGBoost is fitted with a quantile objective to predict the 10th percentile, the median and the 90th percentile. The median is the forecast and the other two give an 80% range. The random forest gives a single forecast.
- **What to predict.** Either the level of admissions in the target week, or the change from this week to the target week. For a tree model the two are not the same. A model that predicts levels can never forecast above the highest level it was trained on; a model that predicts the change can.
- **Feature set.** Lags and season alone, then with Dhaka's recent weeks, with weather, and with both.
- **Settings.** A small grid for each model, because one validation year cannot support a fine search.

Python Notebook 05: the two models and their grids. QUANTILES = [0.1, 0.5, 0.9] (line breaks added)

```
MAKE = {"XGBoost": lambda depth, trees: XGBRegressor(objective="reg:quantileerror", quantile_alpha=QUANTILES,
              max_depth=depth, n_estimators=trees, learning_rate=0.05, subsample=0.8, colsample_bytree=0.8,
              min_child_weight=5, random_state=1),
        "Random forest": lambda leaf, share: RandomForestRegressor(n_estimators=100, min_samples_leaf=leaf,
                          max_features=share, n_jobs=-1, random_state=1)}
GRID = {"XGBoost": {"depth": [2, 3], "trees": [100, 300]}, "Random forest": {"leaf": [2, 5, 10], "share": [0.33, 0.5]}}
```

Table 30. Mean absolute error on 2025 of the best setting for each model, target and feature set (80 candidates in total).

Model	Predicts	Features	Best settings	1 week	2 weeks	3 weeks	4 weeks	Mean
XGBoost	level	lags and season	depth 2, trees 300	44.3	57.0	67.8	76.9	61.5
XGBoost	level	+ Dhaka	depth 2, trees 300	40.1	56.7	68.0	79.0	61.0
XGBoost	level	+ weather	depth 2, trees 300	39.6	52.9	65.7	71.3	57.4
XGBoost	level	+ Dhaka and weather	depth 2, trees 300	39.2	54.6	64.5	73.8	58.0
XGBoost	change	lags and season	depth 3, trees 100	36.7	48.6	61.5	70.3	54.3
XGBoost	change	+ Dhaka	depth 2, trees 100	37.7	49.6	64.2	70.8	55.5
XGBoost	change	+ weather	depth 3, trees 100	36.9	47.4	56.5	63.7	51.1
XGBoost	change	+ Dhaka and weather	depth 3, trees 100	37.0	46.6	58.4	70.4	53.1
Random forest	level	lags and season	leaf 2, share 0.33	40.8	53.1	66.3	69.9	57.5
Random forest	level	+ Dhaka	leaf 2, share 0.33	43.2	53.8	64.8	74.7	59.1
Random forest	level	+ weather	leaf 2, share 0.33	38.1	48.4	59.8	62.9	52.3
Random forest	level	+ Dhaka and weather	leaf 2, share 0.33	39.0	51.9	60.7	66.1	54.5
Random forest	change	lags and season	leaf 10, share 0.33	37.4	55.9	75.0	87.7	64.0
Random forest	change	+ Dhaka	leaf 5, share 0.33	38.0	61.1	83.2	101.5	71.0
Random forest	change	+ weather	leaf 10, share 0.33	40.7	57.9	69.1	85.7	63.4
Random forest	change	+ Dhaka and weather	leaf 10, share 0.33	40.7	62.1	95.1	124.5	80.6

The rule was fixed before looking: the candidate with the lowest mean error wins, and Dhaka or weather features are kept only if they lower the error by more than 2% against lags and season alone. The winners are an XGBoost model that predicts the **change**, with weather features, depth 3 and 100 trees (MAE 51.1), and a random forest that predicts the **level**, with weather features (MAE 52.3). Dhaka's recent weeks did not help either model.

The choice should not be read as sharper than it is. The gap between the best and the fifth best candidate is 2.3 admissions for XGBoost and 3.3 for the random forest, so the choice among the leading candidates is within noise.

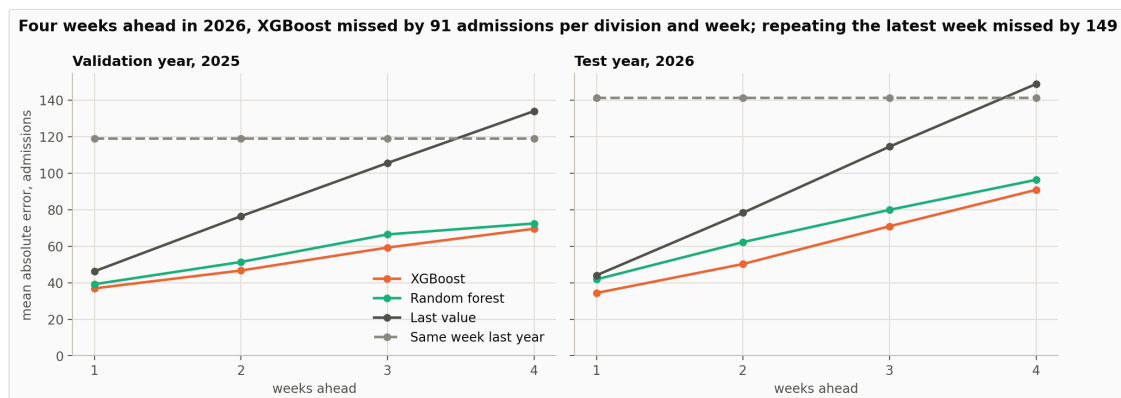
8.3 Rolling forecasts and their accuracy

The models are then used the way they would be used in practice. In every week from 1 December 2024 to 20 September 2026 (95 origin weeks) the model is retrained on everything known up to that week and forecasts the next four weeks for each division. Target, features and settings stay as chosen. This gives 1,664 scored forecasts per model for 2025 and 1,248 for 2026.

Table 31. Mean absolute error by horizon, in admissions per division and week.

Year	Forecast	1 week	2 weeks	3 weeks	4 weeks	All	Better than last value, %	Bias
2025	XGBoost	36.9	46.7	59.3	69.6	53.1	+41.4	-14.3
2025	Random forest	39.2	51.4	66.5	72.5	57.4	+36.7	-10.0
2025	Last value	46.3	76.5	105.7	134.1	90.6	+0.0	+0.5
2025	Same week last year	118.9	118.9	118.9	118.9	118.9	-31.2	-4.4
2026	XGBoost	34.4	50.3	71.0	90.9	61.6	+36.1	-52.9
2026	Random forest	41.8	62.3	80.0	96.4	70.1	+27.3	-61.9
2026	Last value	44.1	78.4	114.6	148.9	96.5	+0.0	-67.2
2026	Same week last year	141.3	141.3	141.3	141.3	141.3	-46.4	-84.6

Bias is the mean of forecast minus actual. Negative means the forecast is too low on average.

**Figure 20.** Mean absolute error against the horizon, validation year (left) and test year (right).

The validation scores are slightly flattering, because features and settings were chosen on that year. The test year is the fair score. In 2026 XGBoost missed by 34.4 admissions one week ahead and by 90.9 four weeks ahead, against 44.1 and 148.9 for repeating the latest week. Over all horizons its error is 36% below the last value, and the random forest's is 27% below.

The bias column is the weak point. In 2026 both models forecast too low on average, XGBoost by 52.9 admissions per division and week. The 2026 season is larger than the two before it, and the models were slow to follow the rise.

Table 32. Mean absolute error in 2026 by division, all horizons together.

Division	Mean weekly admissions	XGBoost	Random forest	Last value	Same week last year	XGBoost better than last value, %
Dhaka	743.1	179.9	216.2	320.3	361.3	43.8
Khulna	301.1	87.2	111.9	118.7	244.9	26.6
Chattogram	272.8	74.3	75.4	112.6	133.8	34.0
Barishal	247.7	51.6	54.0	88.0	167.2	41.4
Rajshahi	113.9	37.5	36.4	53.5	90.5	30.0
Mymensingh	102.3	31.0	34.9	44.6	78.5	30.5
Rangpur	55.5	27.3	28.2	29.0	48.4	6.0
Sylhet	8.6	4.3	3.8	5.2	5.8	17.0

XGBoost beats the last value in all eight divisions in 2026, but not by the same margin: by 44% in Dhaka and by only 6% in Rangpur, where weekly numbers are small and irregular.

8.4 Can the forecast range be trusted?

Table 33. Share of actual values inside, above and below the XGBoost 80% range.

Year	Weeks ahead	Inside the range, %	Above, %	Below, %	Median width, admissions
2025	1	85.8	7.9	6.2	50.5
2025	2	86.3	7.5	6.2	71.0
2025	3	88.2	5.8	6.0	83.4
2025	4	85.3	7.5	7.2	101.3
2026	1	82.7	9.6	7.7	34.6
2026	2	81.4	10.9	7.7	42.1
2026	3	84.0	9.6	6.4	53.8
2026	4	80.4	10.6	9.0	63.0

An honest 80% range holds about 80 in 100 actual values. In 2025 it held 85% to 88%, and in 2026 80% to 84%, so the ranges are about right and a little wide in 2025. In 2026 the misses are not balanced: about 10% of actual values fell above the range and 6% to 9% below. Above is the side that matters for preparedness, because it means more patients than the plan allowed for.

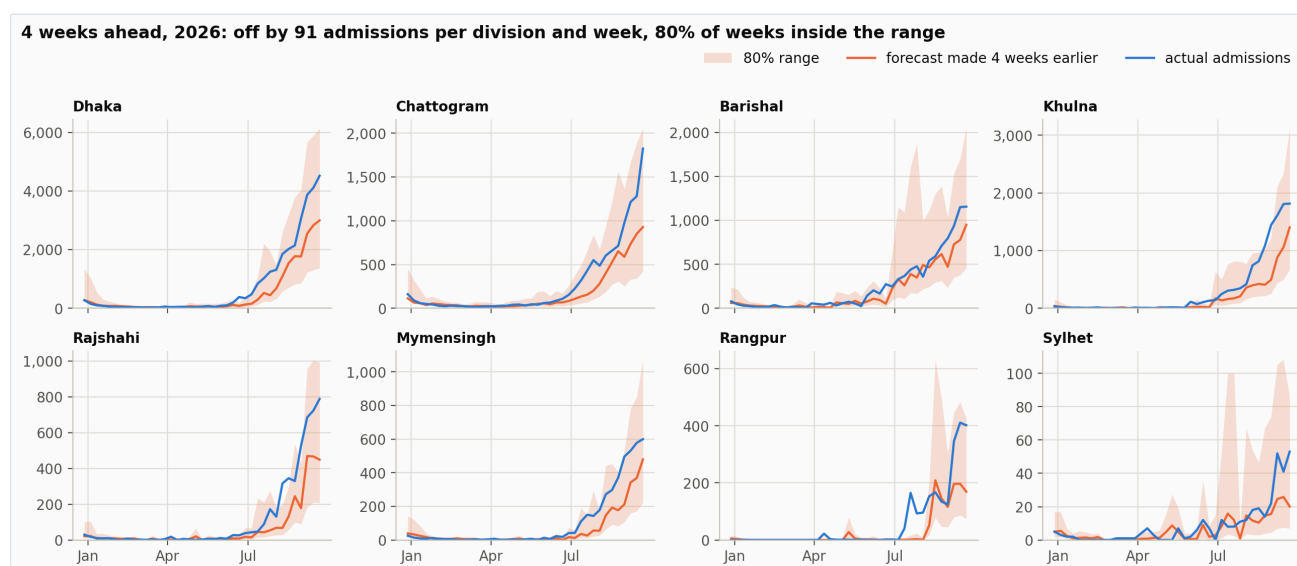


Figure 21. XGBoost four weeks ahead in 2026: actual admissions, the forecast made four weeks earlier and its 80% range, by division.

8.5 What drives the forecasts

SHAP values [15] split each forecast into the contribution of each feature. The chosen model predicts the change from the latest week, so the contributions show what moves the forecast up or down from there.

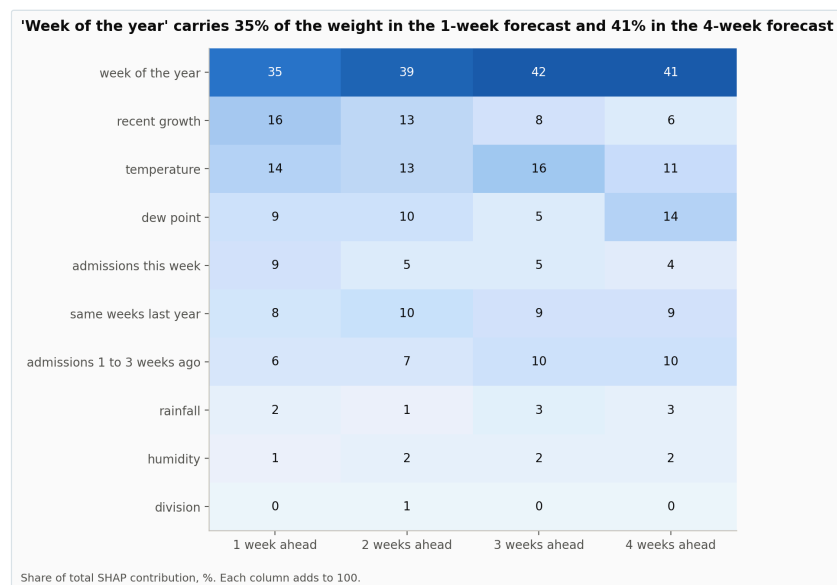


Figure 22. Share of total SHAP contribution by feature group and horizon, XGBoost trained on all the data. Each column adds to 100.

The week of the year carries the most weight: 35% of the total one week ahead and 41% four weeks ahead. Recent growth matters most at short horizons (16% at one week, 6% at four). Temperature and dew point together carry 21% to 25%. Rainfall, humidity and the division carry 3% or less each. This describes what the model relies on. It does not show what causes dengue.

8.6 Did the weather features earn their place?

Weather was kept because it lowered the 2025 error by more than 2%. A gain on the year used for choosing can be luck, so after the test the rolling forecasts were rerun with lags and season only.

Table 34. Mean absolute error with and without the weather features.

Model	2025, with weather	2025, lags and season only	2025 gain, %	2026, with weather	2026, lags and season only	2026 gain, %
XGBoost	53.1	56.5	5.9	61.6	62.1	0.8
Random forest	57.4	60.1	4.6	70.1	70.7	0.8

The gain of 5.9% for XGBoost in 2025 shrank to 0.8% in 2026, and the random forest shows the same pattern. Weather helped on the year it was chosen on and added almost nothing on the test year. This check came after the test and changed nothing in the saved forecasts. I report it so that no claim about weather rests on the validation year alone. It agrees with sections 5 and 7: once a model knows the season and the latest weeks, weather has little left to add at these horizons.

9 Statistical forecasts

R script 08_statistical_forecasts.R

The same forecasts are now made with three classical models. Each is retrained every week on everything known up to that week, exactly as in section 8, and scored on the same list of forecasts. Nothing in this script is tuned on 2025 or 2026, so both years are fair tests for these models.

9.1 The three models

- **ARIMA with Fourier terms.** One model per division on the log scale: a regression on two yearly waves (two pairs of sine and cosine terms) with ARIMA errors [10]. The order is identified once on the training years by the Box-Jenkins steps [16] and then re-estimated every week.
- **ETS.** Exponential smoothing per division on the log scale, with the form chosen automatically every week [17]. It has no seasonal part, because a 52-week season needs more years than the data has.
- **Negative binomial.** One regression for all divisions and one per horizon, on counts: the target week's admissions from the division, the log admissions of the last two weeks and the two yearly waves. Its 80% range comes from the quantiles of the fitted negative binomial distribution.

9.2 Box-Jenkins identification of the ARIMA orders

The model is a regression on yearly waves with ARIMA errors, so every step works on the training series (2023 and 2024) after the waves are taken out.

Step 1. Is differencing needed?

A KPSS statistic above 0.463 rejects a stable level at the 5% level, and the series is then differenced once. The ADF test, which starts from the opposite hypothesis, is the second opinion.

Table 35. Stationarity of each division series after removing the yearly waves (critical value 0.463).

Division	KPSS, level	KPSS, after differencing	d used	d by ADF	ADF agrees
Barishal	0.561	0.134	1	0	no
Chattogram	0.758	0.133	1	1	yes
Dhaka	0.715	0.113	1	1	yes
Khulna	0.430	0.134	0	0	yes
Mymensingh	0.250	0.059	0	0	yes
Rajshahi	0.686	0.145	1	1	yes
Rangpur	0.765	0.078	1	0	no
Sylhet	0.709	0.073	1	1	yes

Six of the eight divisions need one difference. Khulna and Mymensingh need none. The ADF test gives the same answer in six of eight divisions, and where the two disagree (Barishal and Rangpur) the KPSS result is used.

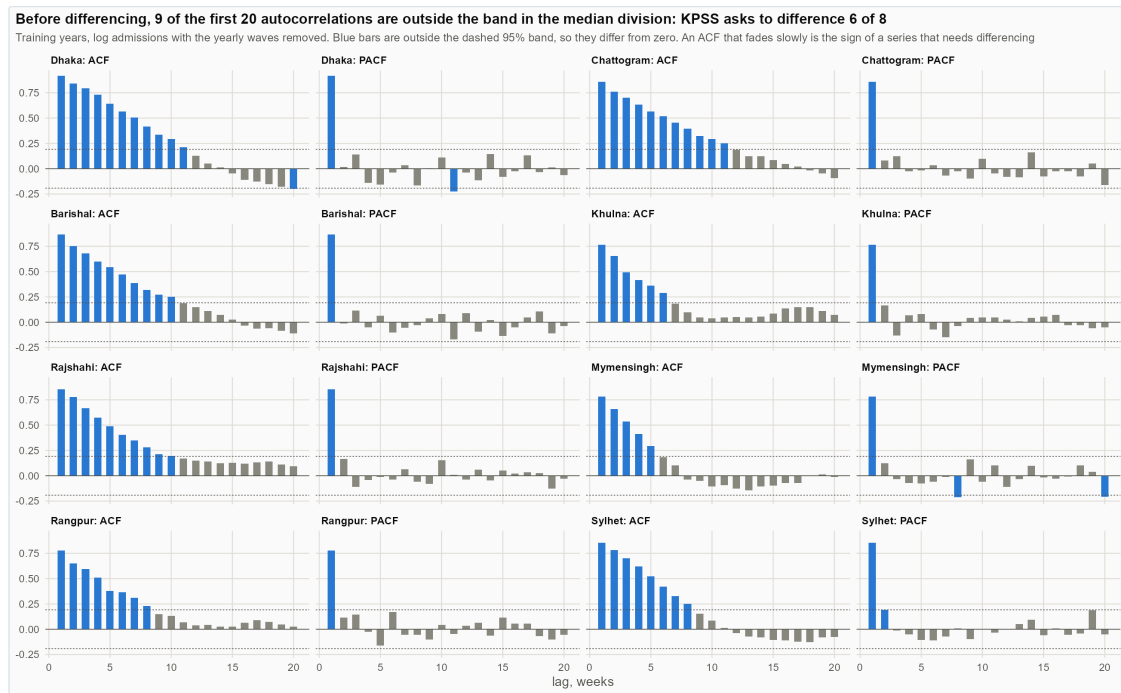


Figure 23. ACF and PACF of each division series before differencing (training years, yearly waves removed). An ACF that fades slowly is the sign of a series that needs differencing.

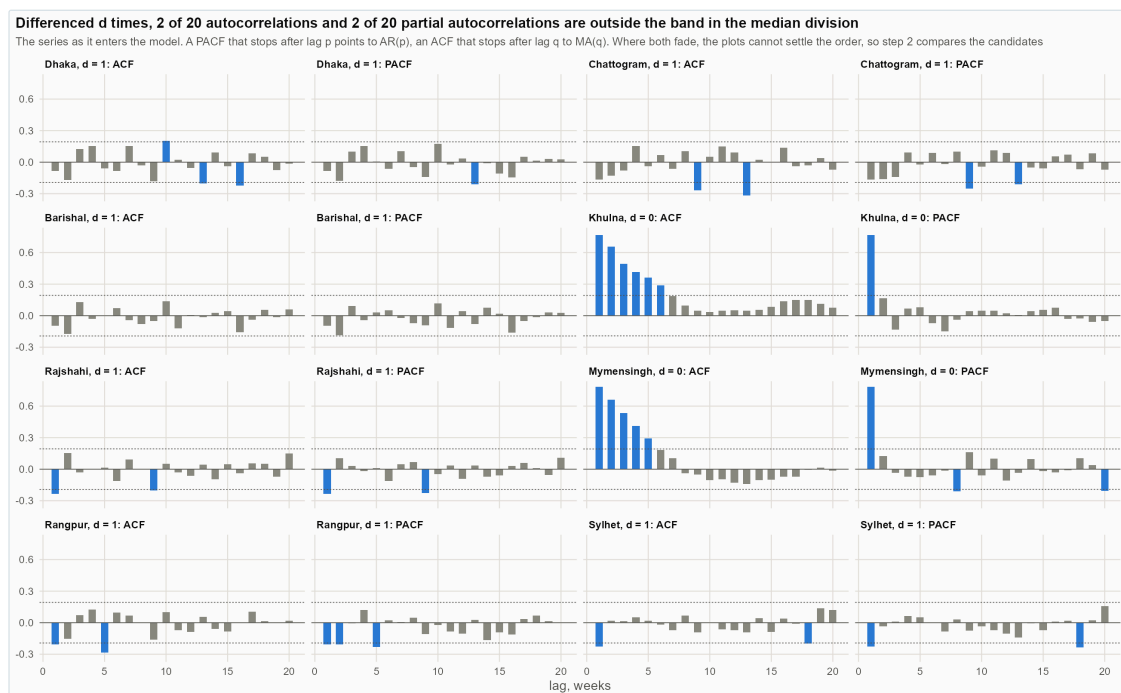


Figure 24. ACF and PACF of each series as it enters the model, differenced d times. Blue bars lie outside the 95% band.

After differencing few bars remain outside the band. In most divisions both the ACF and the PACF fade without a clean cut-off, so the plots alone cannot settle the order. That is what the next step is for.

Step 2. Candidate models and selection

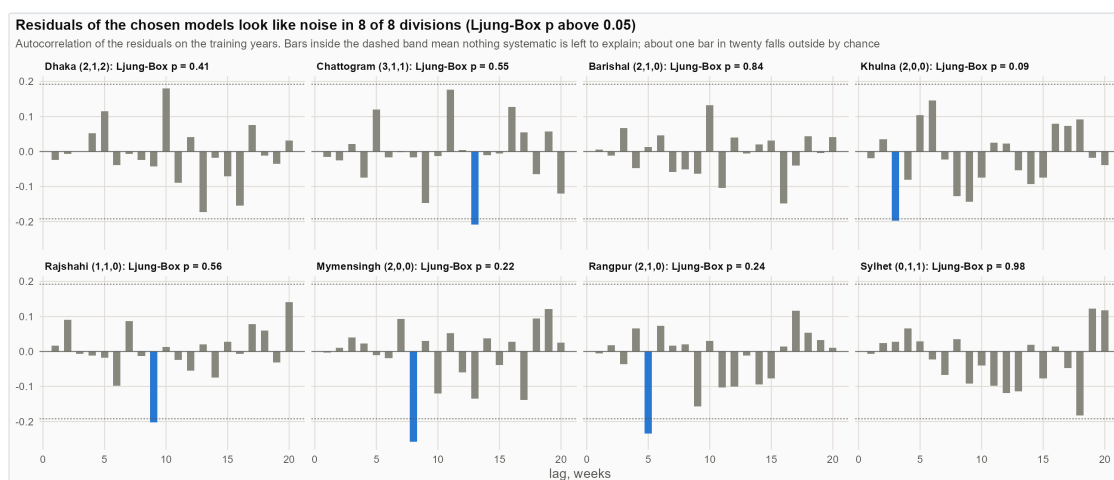
Every order up to $p = 3$ and $q = 3$ is fitted, with and without drift when $d = 1$, and ranked by AICc. A candidate with an AR or MA root on the unit circle is marked inadmissible and cannot be chosen, because that is the sign of an over-fitted model.

Table 36. The ARIMA order used in each division (lowest AICc), what AIC and BIC would have chosen, and the residual check.

Division	Order used (p,d,q)	AIC	AICc	BIC	Best by AIC	Best by BIC	AICc gap of the BIC choice	Ljung-Box p
Barishal	(2,1,0)	163.3	164.5	181.8	(2,1,0)	(0,1,0)	0.9	0.836
Chattogram	(3,1,1)	149.7	151.6	173.4	(3,1,1)	(0,1,1)	2.8	0.550
Dhaka	(2,1,2)	67.5	69.4	91.2	(2,1,2)	(0,1,0)	0.3	0.411
Khulna	(2,0,0)	195.7	197.2	216.8	(1,0,3)	(2,0,0)	0.0	0.091
Mymensingh	(2,0,0)	195.5	197.1	216.7	(2,0,0)	(1,0,0)	0.5	0.219
Rajshahi	(1,1,0)	178.9	179.7	194.7	(1,1,0)	(1,1,0)	0.0	0.565
Rangpur	(2,1,0)	239.4	240.6	257.8	(2,1,0)	(0,1,1)	0.0	0.236
Sylhet	(0,1,1)	202.0	202.9	217.8	(0,1,1)	(0,1,1)	0.0	0.979

The three criteria choose the same order in only two of the eight divisions. BIC prefers a smaller model in five, and in four of those the smaller model is within 2 AICc points of the one used, so the data cannot tell the two apart. I kept the AICc choice throughout as a fixed rule and did not pick per division.

Steps 3 and 4. Estimation and diagnostic checking

**Figure 25.** Autocorrelation of the residuals of the chosen models on the training years, with the Ljung-Box p-value of each division.

The residuals of the chosen models look like noise in all eight divisions: every Ljung-Box p-value is above 0.05, from 0.091 in Khulna to 0.979 in Sylhet.

R Script 08: the yearly waves and the weekly refit of the ARIMA model, with $y = \log(1 + \text{admissions})$ (line breaks added)

```
waves <- function(d) { w <- as.numeric(format(d + 3, "%j")) / 365.25
  cbind(s1 = sin(2 * pi * w), c1 = cos(2 * pi * w), s2 = sin(4 * pi * w), c2 = cos(4 * pi * w)) }

fit_arima <- function(d, o, ord) { X <- waves(d$week_start); Xf <- waves(o + 7 * HORIZONS)
  m <- tryCatch(Arima(d$y, order = c(ord$p, ord$d, ord$q), include.drift = ord$drift, xreg = X, method = "ML"),
    error = function(e) Arima(d$y, order = c(0, 1, 1), xreg = X, method = "ML"))
  forecast(m, xreg = Xf, level = 80) }
```

9.3 Accuracy

Table 37. Mean absolute error by horizon, in admissions per division and week.

Year	Forecast	1 week	2 weeks	3 weeks	4 weeks	All	Better than last value, %	Bias
2025	ARIMA with Fourier terms	56.2	90.8	121.4	153.4	105.5	-16.3	+14.4
2025	ETS	50.2	81.1	114.8	150.1	99.1	-9.3	+35.9
2025	Negative binomial	61.8	117.6	172.3	217.3	142.3	-57.0	+67.9
2025	Last value	46.3	76.5	105.7	134.1	90.6	+0.0	+0.5
2025	Same week last year	118.9	118.9	118.9	118.9	118.9	-31.2	-4.4
2026	ARIMA with Fourier terms	31.1	40.8	48.9	56.3	44.3	+54.1	-11.3
2026	ETS	34.0	49.8	64.3	85.5	58.4	+39.5	-25.3
2026	Negative binomial	38.1	56.2	68.8	81.2	61.1	+36.7	+10.3
2026	Last value	44.1	78.4	114.6	148.9	96.5	+0.0	-67.2
2026	Same week last year	141.3	141.3	141.3	141.3	141.3	-46.4	-84.6

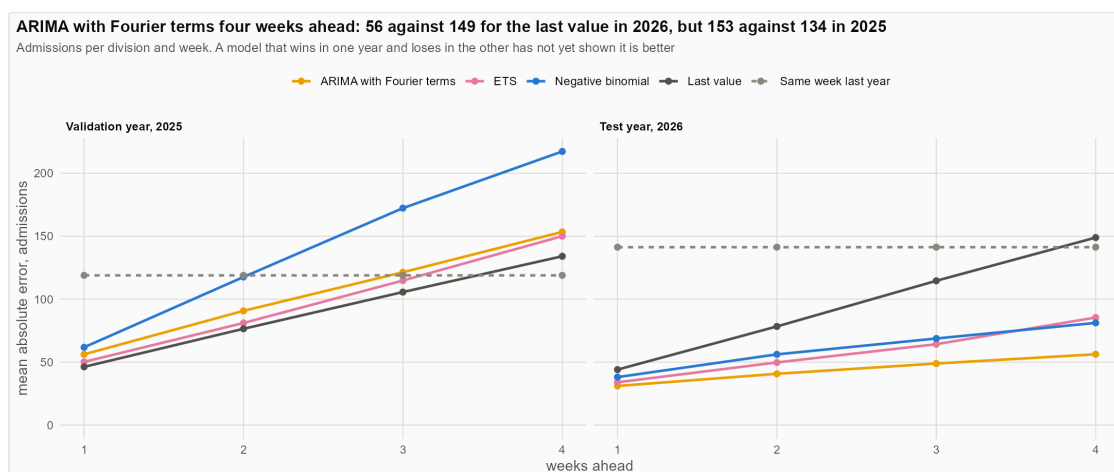


Figure 26. Mean absolute error against the horizon for the statistical models and the two no-model forecasts.

The two years tell opposite stories. In 2026 all three models beat the last value clearly: ARIMA with Fourier terms by 54% over all horizons, ETS by 40% and the negative binomial model by 37%. Four weeks ahead the ARIMA error is 56.3 admissions against 148.9 for the last value. In 2025 all three were worse than the last value, ARIMA by 16%, ETS by 9% and the negative binomial model by 57%.

A model that wins in one year and loses in the other has not yet shown it is better. Section 10 looks at why.

Table 38. Mean absolute error in 2026 by division, all horizons together.

Division	Mean weekly admissions	ARIMA with Fourier terms	ETS	Negative binomial	Last value	Same week last year
Dhaka	743.1	99.1	162.1	132.4	320.3	361.3
Khulna	301.1	55.2	71.1	82.7	118.7	244.9
Chattogram	272.8	63.6	80.5	74.3	112.6	133.8
Barishal	247.7	52.2	62.2	113.3	88.0	167.2
Rajshahi	113.9	22.0	29.4	31.4	53.5	90.5
Mymensingh	102.3	23.4	26.7	23.9	44.6	78.5
Rangpur	55.5	34.9	29.6	26.6	29.0	48.4
Sylhet	8.6	3.6	5.3	3.9	5.2	5.8

In 2026 the ARIMA model beats the last value in seven of eight divisions. The exception is Rangpur (34.9 against 29.0).

9.4 Ranges

Table 39. Share of actual values inside the 80% range, %.

Year	Weeks ahead	ARIMA with Fourier terms	ETS	Negative binomial
2025	1	88.0	88.5	91.6
2025	2	85.3	85.8	92.1
2025	3	84.4	86.3	89.4
2025	4	84.1	85.8	88.2
2026	1	83.0	83.3	88.1
2026	2	82.4	82.1	89.7
2026	3	83.7	83.7	89.1
2026	4	84.3	81.4	87.5

The ranges of ARIMA and ETS hold 81% to 89% of actual values, close to the 80% they aim for. The negative binomial ranges are wider than needed, holding 88% to 92%.

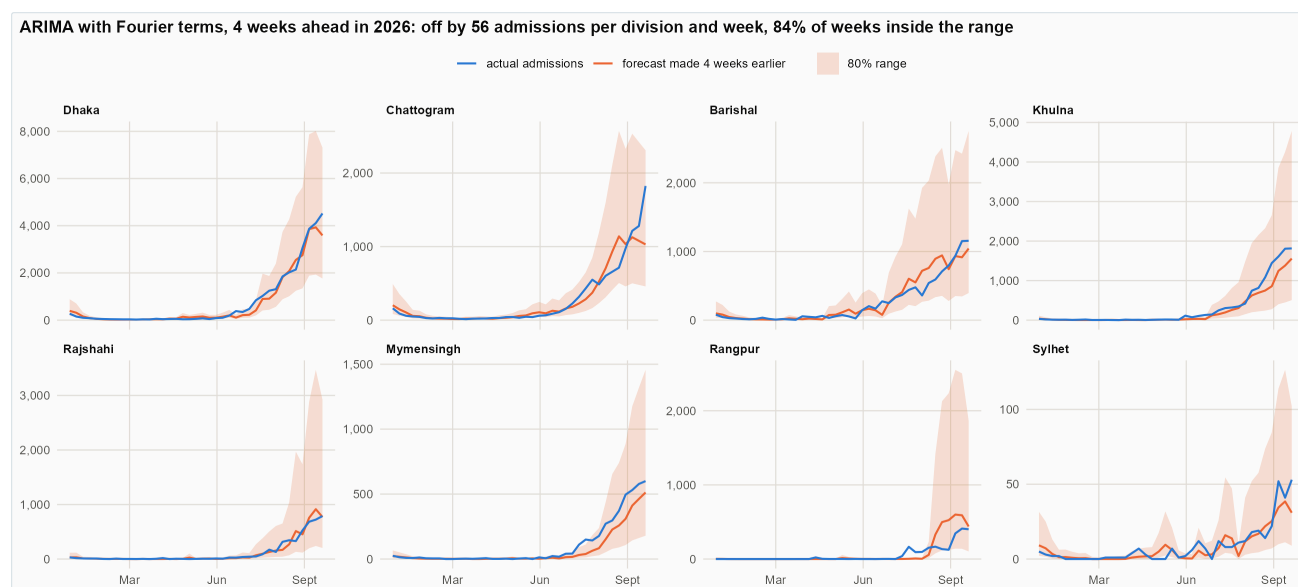


Figure 27. ARIMA with Fourier terms four weeks ahead in 2026: actual admissions, the forecast made four weeks earlier and its 80% range, by division.

10 Model comparison

R script 09_model_comparison_powerbi.R, part A

No model is fitted in this script. It puts every forecast side by side on exactly the same 2,912 forecasts (8 divisions, 1 to 4 weeks ahead, 1,664 in 2025 and 1,248 in 2026) and asks which model is best and whether the difference is real or luck.

10.1 A simple ensemble and one score for forecast and range

Two things are added before any score is looked at. The first is an **ensemble**: for each forecast, the median of the five models. Its 80% range is the median of the four ranges available, since the random forest gives none. It was defined before the comparison, so it was not chosen to look good.

The second is the **weighted interval score** [18], one number for the forecast and its range together. For an 80% range it adds half the absolute error, a tenth of the width of the range, and a penalty when the actual value falls outside. Lower is better, and it rewards ranges that are narrow but still contain the value.

R Script 09: the ensemble and the weighted interval score (line breaks added)

```
ensemble <- members %>% filter(model %in% MODELS[1:5]) %>% group_by(across(all_of(KEY))) %>%
  summarise(forecast = median(forecast), lower_80 = median(lower_80, na.rm = TRUE),
            upper_80 = median(upper_80, na.rm = TRUE), members = n(), .groups = "drop")

wis = (0.5 * abs_error + 0.1 * ((upper_80 - lower_80) + 10 * pmax(lower_80 - actual, 0)
  + 10 * pmax(actual - upper_80, 0))) / 1.5
```

10.2 Which model is best?

Table 40. All eight forecasts in both years, over all four horizons, ordered by the mean of the two years.

Forecast	MAE 2025	Rank 2025	Better than last value 2025, %	MAE 2026	Rank 2026	Better than last value 2026, %	Mean of the two years
Ensemble (median)	60.7	3	+33.1	49.3	2	+48.9	55.0
XGBoost	53.1	1	+41.4	61.6	5	+36.1	57.4
Random forest	57.4	2	+36.7	70.1	6	+27.3	63.8
ARIMA with Fourier terms	105.4	6	-16.3	44.3	1	+54.1	74.9
ETS	99.1	5	-9.3	58.4	3	+39.5	78.7
Last value	90.6	4	+0.0	96.5	7	+0.0	93.6
Negative binomial	142.3	8	-57.0	61.1	4	+36.7	101.7
Same week last year	118.9	7	-31.2	141.3	8	-46.4	130.1

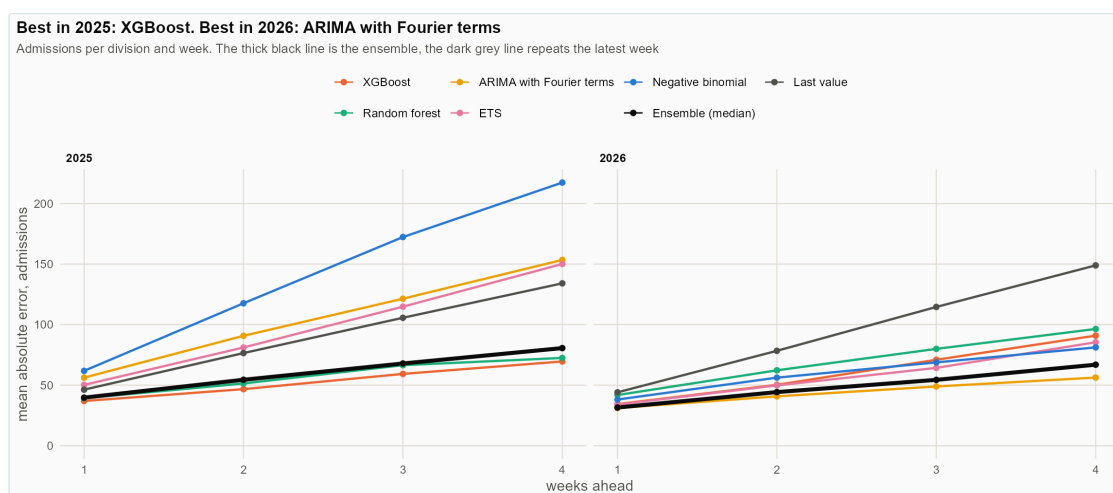


Figure 28. Mean absolute error against the horizon for all models in 2025 and 2026. The thick black line is the ensemble.

No model is best in both years. In 2025 the two machine learning models lead (XGBoost 53.1, random forest 57.4) and all three statistical models are worse than repeating the latest week. In 2026 the order turns over: ARIMA with Fourier terms is best at 44.3, and XGBoost and the random forest fall to fifth and sixth.

Three forecasts beat the last value in both years: the ensemble, XGBoost and the random forest. The ensemble is never the best but is third in 2025 and second in 2026, and it has the lowest error averaged over the two years (55.0). That is the usual experience with forecast combinations, and it was also the experience of the multi-team dengue forecasting challenge reported by Johansson and colleagues [19].

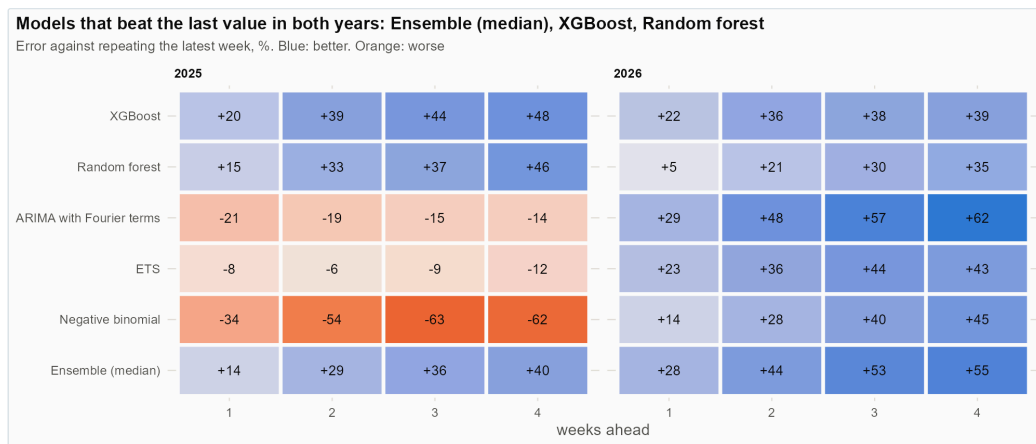


Figure 29. Error against repeating the latest week, in %, by model, year and horizon. Blue is better than the last value, orange is worse.

10.3 Is the difference real?

The Diebold-Mariano test [20] asks whether two forecasts' errors differ by more than week-to-week noise. Errors are averaged over the eight divisions each week, and the test allows for the errors of overlapping forecasts being related, with the small-sample correction of Harvey, Leybourne and Newbold [21].

Table 41. Diebold-Mariano tests over all horizons. A negative difference means the first forecast has the smaller error.

Year	Forecast	Against	Weeks	Mean difference in error	t	p-value	Verdict
2025	XGBoost	Last value	55	-47.4	-2.26	0.028	XGBoost better
2026	XGBoost	Last value	42	-45.4	-2.32	0.025	XGBoost better
2025	Random forest	Last value	55	-43.5	-1.94	0.058	no clear difference
2026	Random forest	Last value	42	-33.9	-1.91	0.064	no clear difference
2025	ARIMA with Fourier terms	Last value	55	+2.8	0.08	0.940	no clear difference
2026	ARIMA with Fourier terms	Last value	42	-61.0	-2.42	0.020	ARIMA with Fourier terms better
2025	ETS	Last value	55	+4.9	0.34	0.730	no clear difference
2026	ETS	Last value	42	-39.6	-2.36	0.023	ETS better
2025	Negative binomial	Last value	55	+36.8	0.73	0.470	no clear difference
2026	Negative binomial	Last value	42	-44.2	-2.08	0.044	Negative binomial better
2025	Ensemble (median)	Last value	55	-40.1	-1.69	0.097	no clear difference
2026	Ensemble (median)	Last value	42	-57.2	-2.64	0.012	Ensemble (median) better
2025	XGBoost	ARIMA with Fourier terms	55	-50.3	-1.98	0.053	no clear difference
2026	XGBoost	ARIMA with Fourier terms	42	+15.5	1.46	0.150	no clear difference
2025	Ensemble (median)	XGBoost	55	+7.3	1.12	0.270	no clear difference
2026	Ensemble (median)	XGBoost	42	-11.8	-2.12	0.040	Ensemble (median) better
2025	Ensemble (median)	ARIMA with Fourier terms	55	-43.0	-2.15	0.036	Ensemble (median) better
2026	Ensemble (median)	ARIMA with Fourier terms	42	+3.8	0.59	0.560	no clear difference

Only XGBoost is clearly better than the last value in both years ($p = 0.028$ in 2025 and 0.025 in 2026). In 2026 ARIMA, ETS, the negative binomial model and the ensemble are also clearly better, but none of them was in 2025. The random forest is close in both years and clear in neither ($p = 0.058$ and 0.064).

Head to head, XGBoost and ARIMA cannot be separated in either year ($p = 0.053$ and 0.15). The ensemble is clearly better than ARIMA in 2025 and than XGBoost in 2026. Many tests are run here, so a single p -value just under 0.05 is weak evidence. A result that holds in both years is worth more.

10.4 Ranges

Table 42. Error, bias, range coverage and weighted interval score over all horizons.

Year	Forecast	MAE	Bias	Inside the 80% range, %	Weighted interval score	Rank by MAE
2025	XGBoost	53.1	-14.3	86.9	43.0	1
2025	Random forest	57.4	-10.0			2
2025	ARIMA with Fourier terms	105.4	+14.4	85.6	70.4	6
2025	ETS	99.1	+35.9	86.9	80.1	5
2025	Negative binomial	142.3	+67.9	90.3	97.8	8
2025	Ensemble (median)	60.7	-3.7	90.4	53.4	3
2025	Last value	90.6	+0.5			4
2025	Same week last year	118.9	-4.4			7
2026	XGBoost	61.6	-52.9	82.5	39.2	5
2026	Random forest	70.1	-61.9			6
2026	ARIMA with Fourier terms	44.3	-11.3	84.1	41.5	1
2026	ETS	58.4	-25.3	82.9	52.6	3
2026	Negative binomial	61.1	+10.3	88.6	56.1	4
2026	Ensemble (median)	49.3	-34.3	85.9	43.0	2
2026	Last value	96.5	-67.2			7
2026	Same week last year	141.3	-84.6			8

The random forest and the two no-model forecasts give no range, so they have no coverage or interval score.

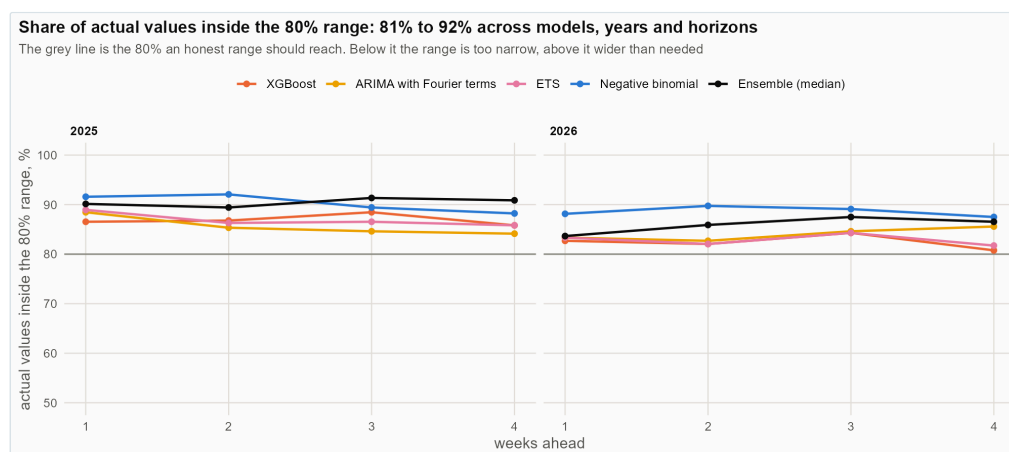


Figure 30. Share of actual values inside the 80% range by model, year and horizon. The grey line is the 80% an honest range should reach.

All five ranges hold between 81% and 92% of actual values, so none of them gives false confidence. XGBoost has the lowest weighted interval score in both years (43.0 and 39.2), even in 2026 when its point error is only fifth, because its ranges are narrower and still hold 82.5% of the values. The bias column shows the shared weakness of 2026: every forecast except the negative binomial model was too low on average, the tree models most of all.

10.5 Did the models see the 2025 peak coming?

2025 is the only complete season inside the forecast period. For each division the week with the highest forecast is compared with the week with the highest actual value, and the size of the forecast peak with the actual one.

Table 43. The forecast 2025 peak against the actual one: median over the eight divisions.

Forecast	Weeks late, 1 week ahead	Size ratio, 1 week ahead	Weeks late, 4 weeks ahead	Size ratio, 4 weeks ahead
XGBoost	+1.0	0.95	+1.0	0.81
Random forest	+1.0	0.94	0	0.98
ARIMA with Fourier terms	-0.5	0.77	-10.0	1.27
ETS	+1.0	1.01	+3.0	1.31
Negative binomial	-0.5	0.78	-9.5	1.34
Ensemble (median)	+1.0	0.92	-2.0	0.84
Last value	+1.0	1.00	+4.0	1.00
Same week last year	0	1.14	0	1.14

Weeks late: negative means the forecast peak came before the actual peak. Size ratio: forecast peak over actual peak.

This table is the most likely explanation for the turn between the two years. Four weeks ahead, the two models that carry a fixed yearly shape, ARIMA with Fourier terms and the negative binomial model, placed the 2025 peak about ten weeks too early and 27% to 34% too high. The 2025 season was unusual: it started early, rose slowly for 20 weeks and peaked in November. A fixed seasonal shape learned from 2023 and 2024 did not fit it. The tree models, which lean on recent weeks and recent growth, were within a week of the peak but 2% to 19% too low. In 2026 the rise so far has followed a more usual course, which suits the seasonal models. This is a reading of two years, not a tested result.

10.6 The next four weeks

The last forecasts were made in the week of 20 September 2026 for the four weeks after it. They have not happened yet at the time of the data snapshot, and they can be checked against the DGHS dashboard as the weeks pass.

Table 44. National admissions expected in the next four weeks (sum of the division forecasts). The latest actual week had 11,157.

Forecast	Week of 27 Sep	Week of 4 Oct	Week of 11 Oct	Week of 18 Oct
XGBoost	11,695	12,949	13,248	13,638
Random forest	11,695	12,038	11,683	10,537
ARIMA with Fourier terms	11,692	12,100	11,881	11,085
ETS	12,312	13,424	14,498	15,529
Negative binomial	11,040	10,536	10,038	9,161
Ensemble (median)	11,612	12,673	13,097	12,499
Last value	11,157	11,157	11,157	11,157
Same week last year	3,656	4,741	5,293	5,398

Table 45. Ensemble forecast by division.

Division	Week of 27 Sep	Week of 4 Oct	Week of 11 Oct	Week of 18 Oct	80% range, week of 18 Oct
Dhaka	4,724	5,234	5,367	5,375	2,488 to 11,074
Chattogram	1,819	1,914	2,083	1,576	743 to 4,144
Barishal	1,248	1,350	1,361	1,362	403 to 3,282
Khulna	1,867	2,056	2,097	2,052	500 to 4,109
Rajshahi	861	963	993	983	226 to 2,529
Mymensingh	620	658	705	690	164 to 1,304
Rangpur	419	443	435	401	81 to 1,294
Sylhet	54	56	55	60	16 to 160

The ensemble expects 11,612 admissions in the week of 27 September, 12,673 and 13,097 in the two weeks after, and 12,499 in the week of 18 October, about 49,900 over the four weeks. The five models agree closely for the first week (11,040 to 12,312) and spread out after that. By the fourth week they run from 9,161 (negative binomial) to 15,529 (ETS). They also disagree on direction: ETS and XGBoost keep rising, while ARIMA, the random forest and the negative binomial model expect admissions to level off or fall during October. The honest summary for a planner is that the ensemble expects admissions to stay above the current level of about 11,000 a week through mid October, and that the direction after the first two weeks is not settled.

11 Power BI dashboard

R script 09_model_comparison_powerbi.R, part B, and Dengue Dashboard.pbix

The second half of script 09 writes every result of the project as tidy tables for Power BI: 47 CSV files, made up of 9 dimension tables, 5 fact tables, 22 tables of analysis results, 10 tables of forecast evaluation and a glossary. It also writes a table dictionary and a list of 55 relationships, all many to one and single direction, each checked so that every key on the many side exists on the one side.

The rule for the dashboard is that Power BI only displays these numbers. Nothing statistical is recalculated in DAX, so the dashboard cannot drift away from the notebooks and scripts. The findings printed by each notebook and script are exported word for word and shown on the pages.

The report has eight pages, one for each group of questions, a division profile that opens by drill-through from any division, and a tooltip that explains each measure from the glossary. Slicers sit at the top right of the pages that need them, with a reset button, and the page navigation is on the left.

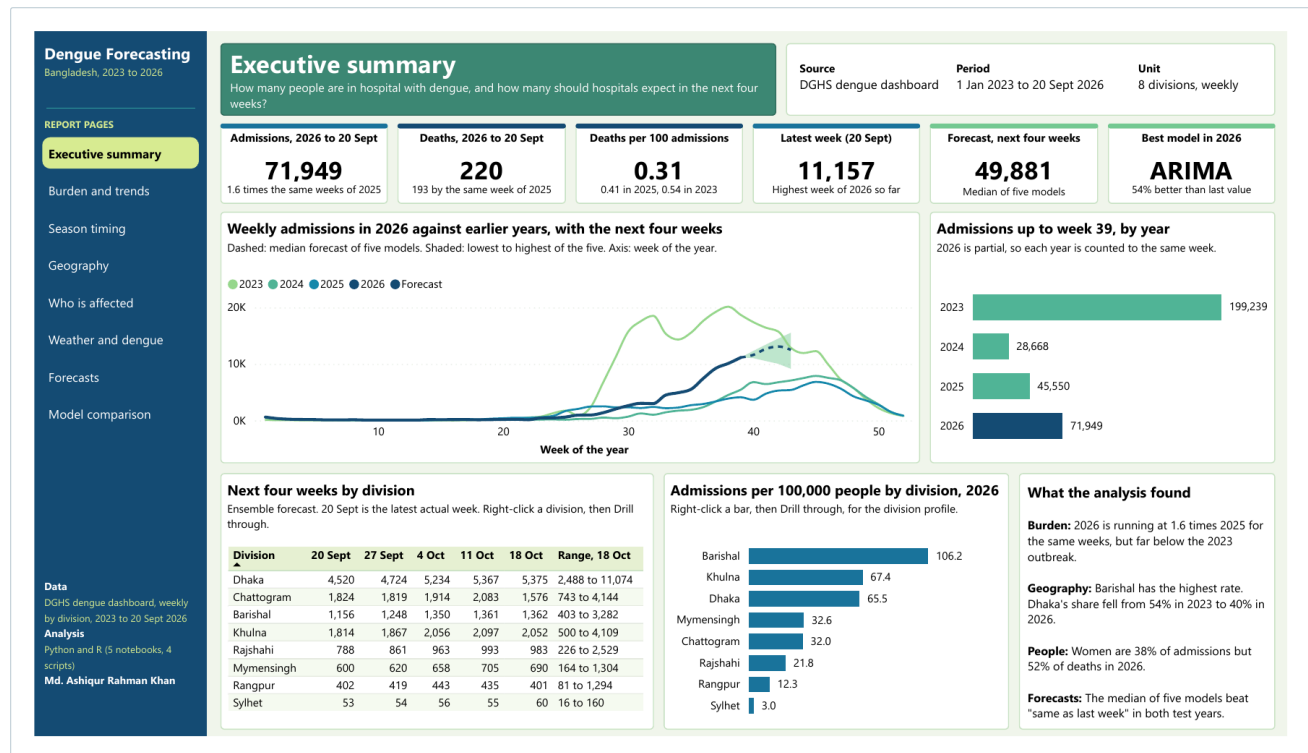


Figure 31. Dashboard page 1, Executive summary. The headline numbers of 2026, the year against earlier years, the forecast for the next four weeks and what the analysis found.

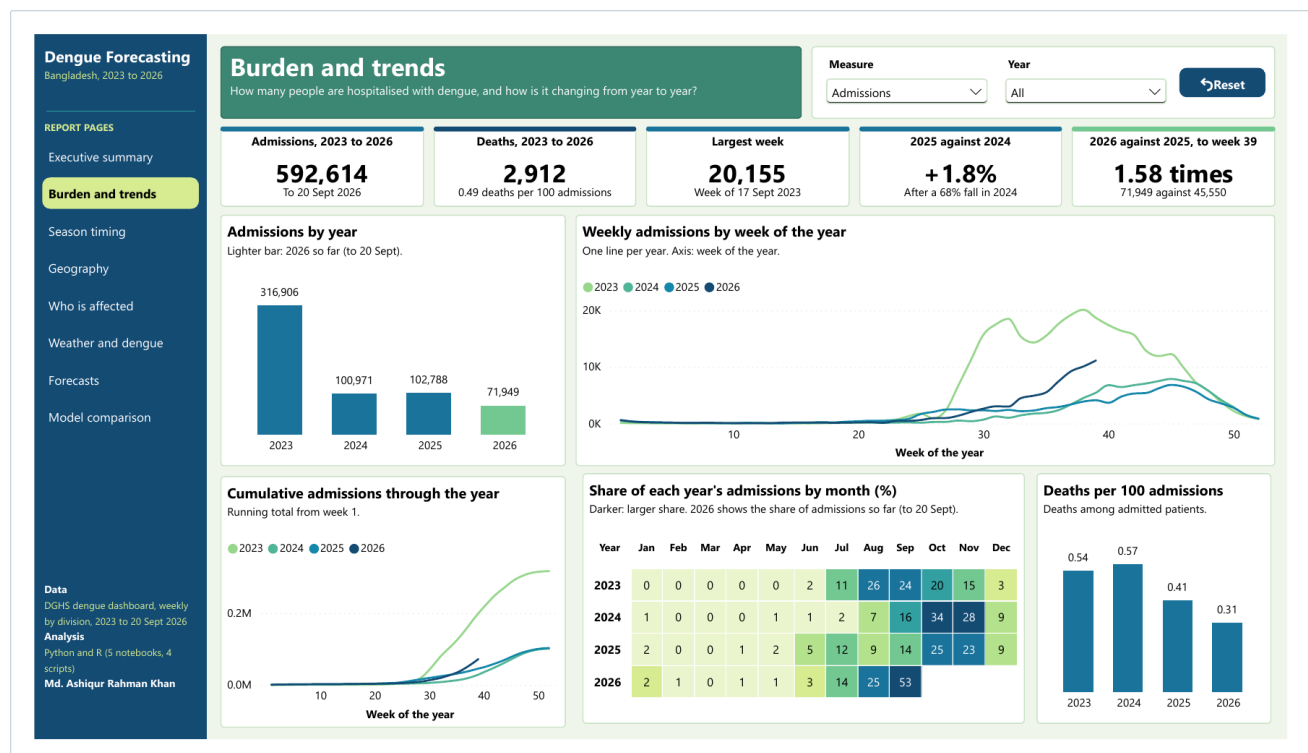


Figure 32. Dashboard page 2, Burden and trends. Admissions and deaths by year, each year by week, cumulative curves, the monthly pattern and deaths per 100 admissions.

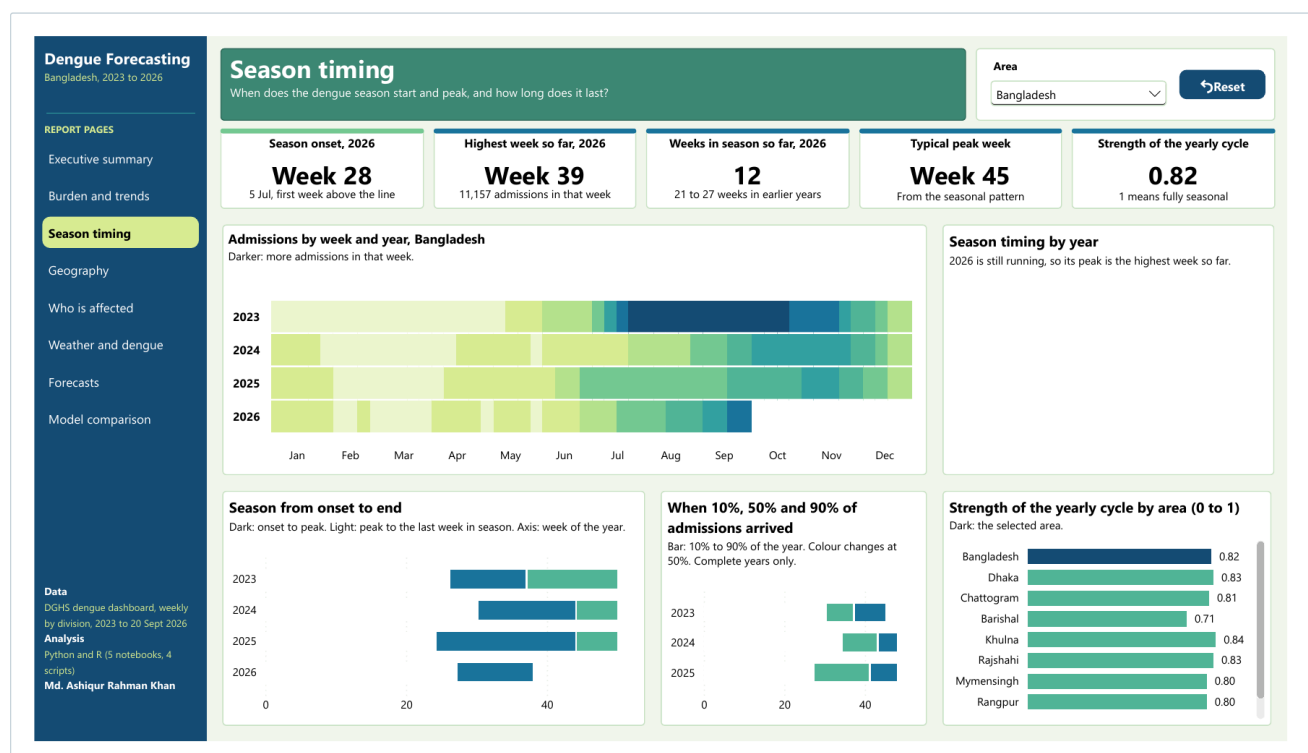


Figure 33. Dashboard page 3, Season timing. Onset, peak and length of the season by year and area, the cumulative timing and the strength of the yearly cycle.

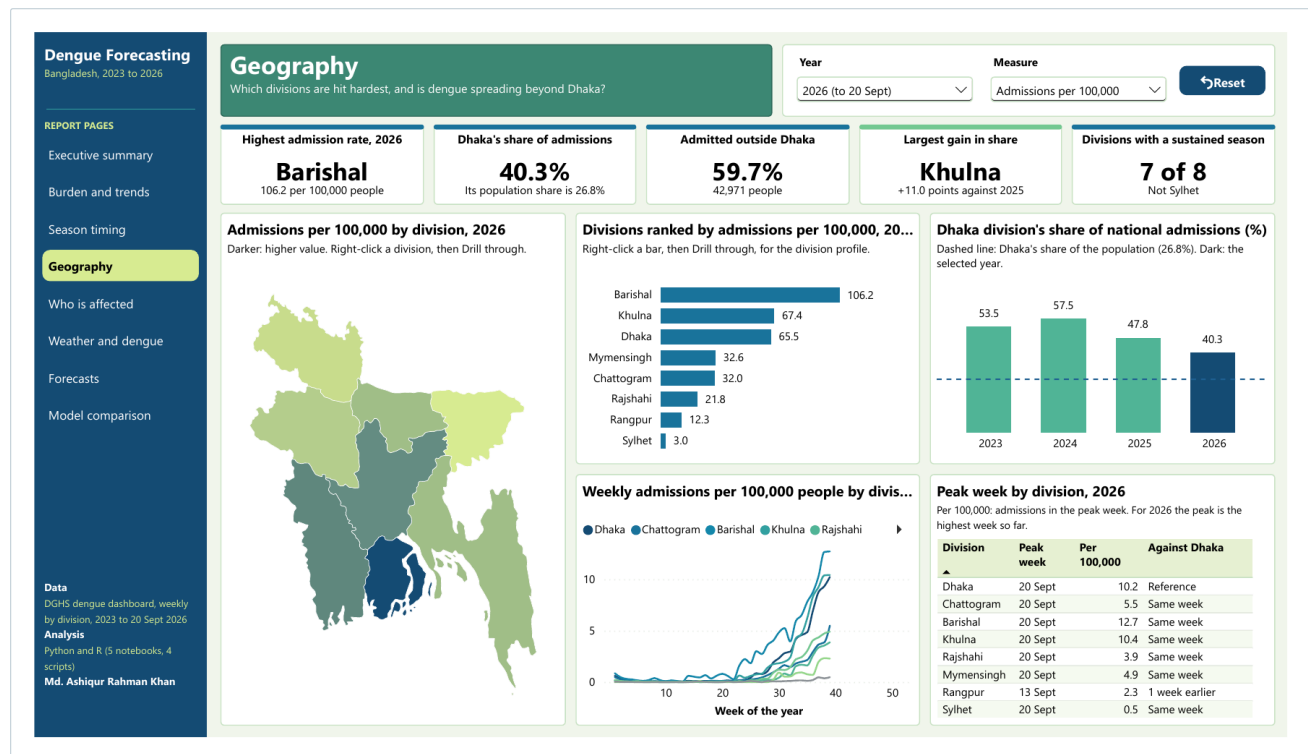


Figure 34. Dashboard page 4, Geography. Admissions per 100,000 by division on a map, the ranking, Dhaka's share against its population share and the peak week by division.

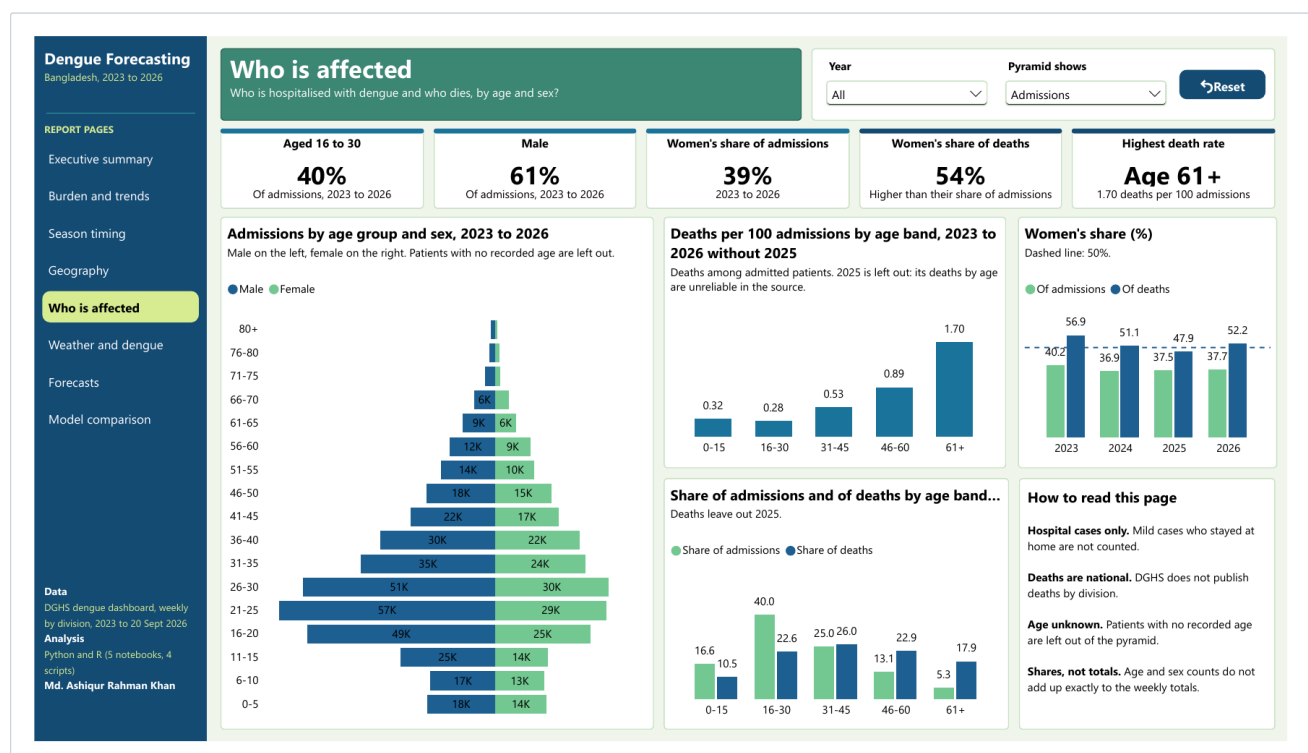


Figure 35. Dashboard page 5, Who is affected. The age and sex pyramid, deaths per 100 admissions by age band and women's share of admissions and of deaths.

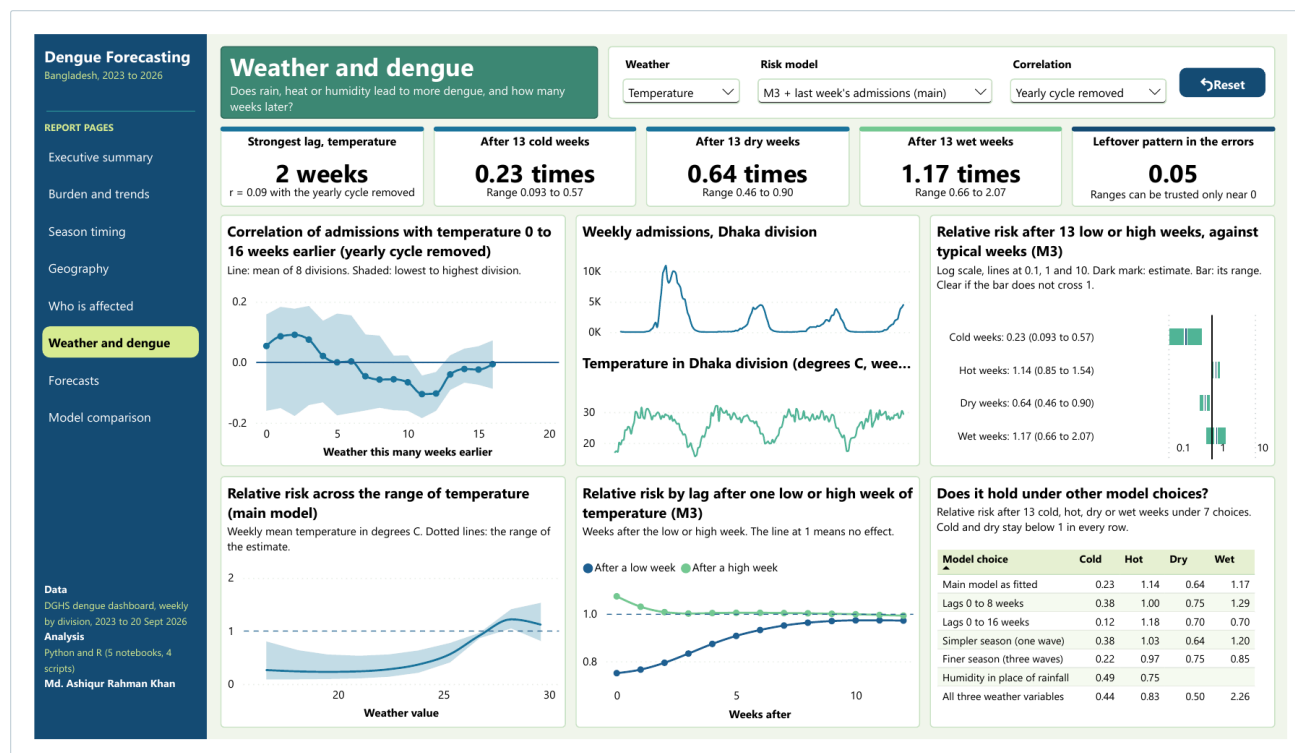


Figure 36. Dashboard page 6, Weather and dengue. Lag correlations, relative risks from the three weather models, the lag curves and the sensitivity table.

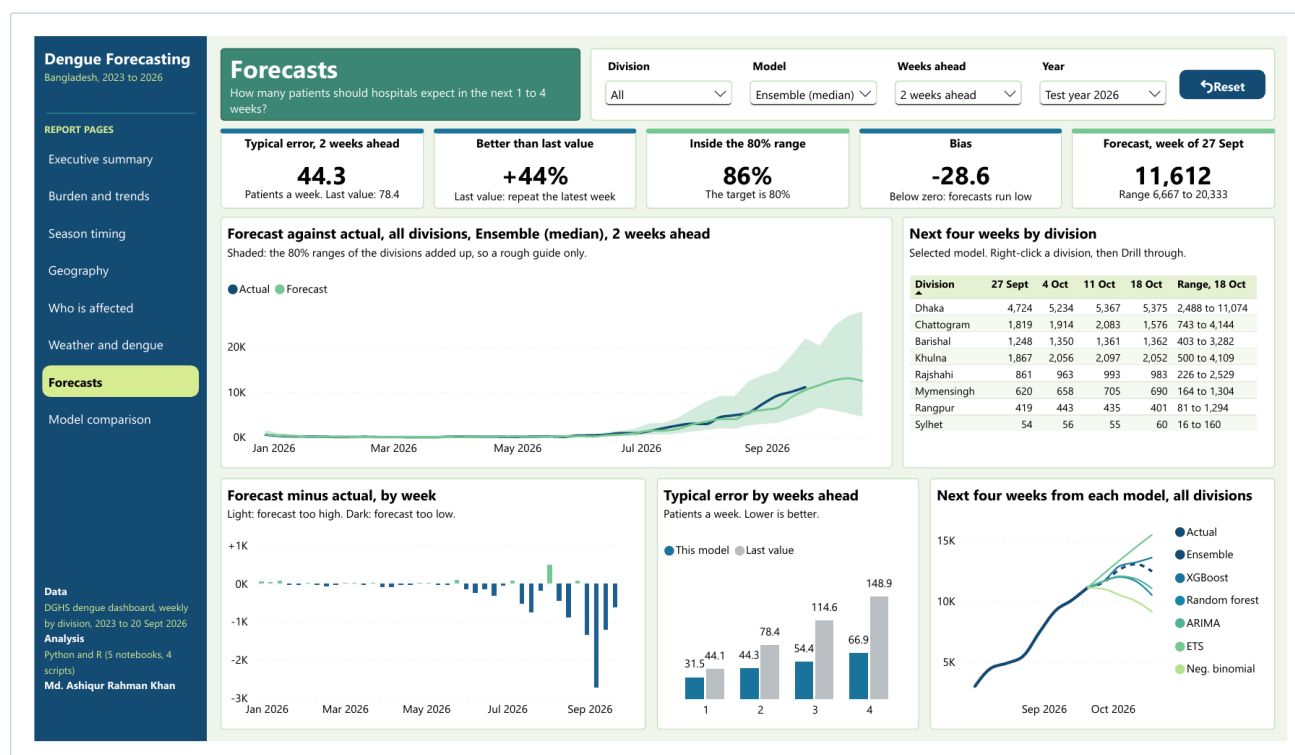


Figure 37. Dashboard page 7, Forecasts. Forecast against actual for any model, division and horizon, the error by horizon and the next four weeks by division.

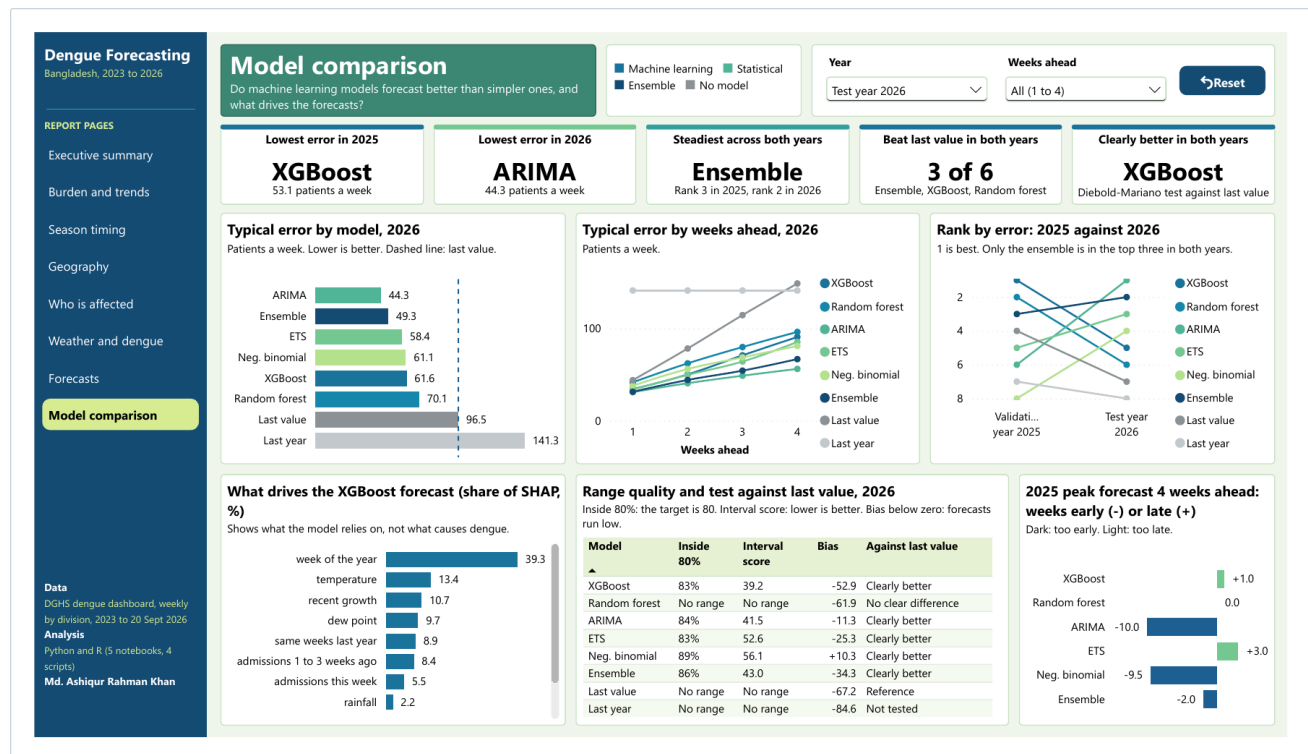


Figure 38. Dashboard page 8, Model comparison. Error and rank of every model in both years, SHAP shares, range quality and the 2025 peak forecast.

12 Discussion

12.1 Answers to the seven questions

Q1. Burden. 2023 was about three times the size of the two years after it, with 316,906 admissions against roughly 101,000 and 103,000. 2026 is again larger: at week 39 it stands at 1.58 times 2025 and 2.5 times 2024. In 2024 and 2025 well under half of the year's admissions had arrived by that week, so the final size of 2026 is still open.

Q2. Season. Dengue admissions in Bangladesh follow one strong yearly cycle, with a seasonal peak about 160 times the seasonal low. The season started between mid June and late July in all four years. The peak moved much more than the onset: mid September in 2023, early November in 2024 and 2025. Only one statement about timing is firm across thresholds, that the 2024 season started latest.

Q3. Geography. Per person, Barishal is hit hardest, not Dhaka. Dhaka division still has more admissions than its share of the population would explain, but its share of the national total has fallen in each of the last two years, while Khulna's has risen sharply in 2026. The idea that other divisions follow Dhaka by some weeks is not supported: on the threshold-free measure the median delay is zero.

Q4. People. Admissions are concentrated in young adults and in men. Deaths are not: women are a minority of admissions and about half of deaths, and the death rate among admitted patients rises several times over from the youngest to the oldest band. The data cannot say why. Later arrival at hospital, differences in who gets admitted, and differences in the disease itself would all produce this pattern.

Q5. Weather. Weather and dengue rise and fall together because both follow the monsoon. Once the season is in the model, what remains is modest: fewer admissions after cooler and drier weather than usual, with the largest single-week effect about five to six weeks later, and no firm sign that hot or very wet weeks add admissions. As a forecasting input at one to four weeks, weather added almost nothing in the test year.

Q6. Forecasts. Useful forecasts are possible at these horizons. In 2026 the best single model missed by 31 admissions per division and week one week ahead and by 56 four weeks ahead, against 44 and 149 for repeating the latest week. The ensemble was close behind and steadier across the two years.

Q7. Machine learning against simpler models. The answer is mixed. XGBoost was the best model in 2025 and the only one that beat the no-model forecast clearly in both years, but in 2026 a per-division ARIMA model with two yearly waves had a lower error, and head to head the two cannot be separated in either year. What the forecasts rely on is plain from the SHAP shares: the time of year first, then the latest weeks and their growth.

12.2 What a hospital planner can take from this

- **Use the ensemble, and read its range.** No single model was reliable across both years. The median of the five was never the best and never poor, and its 80% range held 90% of actual values in 2025 and 86% in 2026.
- **Plan for the upper end.** In 2026 most forecasts were too low on average while cases were rising. When the forecast is rising, the upper end of the range is the safer planning number.
- **Do not use last year as a guide.** Repeating the same week of last year was the worst or second worst forecast in both years, because the seasons differ so much in size.
- **Watch Barishal and Khulna as well as Dhaka.** The burden per person is highest outside Dhaka, and the division shares are moving.

12.3 What I would do next

Three additions would make the forecasts more useful. The first is to score the forecasts for the rest of the 2026 season when the data is in, since the test so far covers the rise and not the peak. The second is to allow for reporting delays, because the latest week on the dashboard can still be revised upward. The third is to extend the series back before 2023 from the archived press releases, so that the models have more than three seasons to learn from.

13 Limitations

These are the limits I think a reader should hold against every number in this report.

- **Hospital admissions, not infections.** The data counts people admitted to the hospitals that report to DGHS. Mild cases, people treated at home and facilities outside the reporting system are not in it. A rise can reflect more reporting as well as more disease.
- **A single source that revises itself.** Everything rests on the dashboard snapshot of 1 October 2026. The dashboard can revise past numbers, and its 2023 total differs slightly from the press release of the time. Rerunning notebook 01 on a later date can change the numbers.
- **Estimated values in the division series.** The 2023 values of Dhaka and Barishal from mid July were rebuilt from press releases, and 119 of the 1,560 division-weeks carry an adjusted value. The national series is not affected. Division-level results for 2023 should be read with this in mind.
- **Missing report days.** 27 days have no report, 19 of them in early 2023, when cases were few. Weekly totals for those weeks are a little low.
- **Deaths by age in 2025.** The age pattern of the 2025 deaths in the source looks wrong and could not be confirmed from an independent source. No conclusion uses it.
- **Three seasons.** The models learned from 2023, 2024 and 2025, and 2023 was about three times the size of the others. A season unlike all three would be forecast poorly. The seasonal findings are descriptions of three complete years, not trends.
- **The test year is not a full season.** The 2026 data ends while cases are still rising. The test covers the quiet months and the rise, not the peak and the decline, and the model ranking could change again once the season is complete.
- **One validation year.** Features and settings of the machine learning models were chosen on 2025 alone, and the leading candidates were within a few admissions of each other. The ARIMA orders were fixed on two training years, and for several divisions two orders fit equally well.
- **Mean absolute error favours large divisions.** The error is measured in admissions, so Dhaka weighs most in every pooled score. The tables by division are the place to check a small division.
- **Many tests.** Ninety Diebold-Mariano tests were run without a correction for multiple testing. Single p-values just under 0.05 are weak evidence.
- **The ensemble range is a simple device.** It is the median of the member ranges, not a properly combined distribution. Its coverage was good in both years, but it has no theory behind it.
- **Weather is a proxy and the analysis is ecological.** One reanalysis grid point at the divisional headquarters stands for a whole division. The weather results are associations at the level of divisions and weeks. They are not causal estimates, and the model does not include mosquito control, serotype changes or population immunity.
- **Population is fixed at the 2022 census.** Rates per 100,000 ignore population growth over the four years.
- **Reporting delays are not modelled.** A live forecast would start from a latest week that may still be revised upward.

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A Reproducing the analysis

The project runs from raw downloads to the dashboard tables without manual steps in between. The Python notebooks were run in Google Colab with the project folder on Google Drive, and the R scripts in RStudio on a copy of the same folder. Each R script sets the project root in one line at the top.

```
Bangladesh Dengue Forecasting Project/
├── data/
│   ├── raw/           dghs_dashboard/ (HTML and JSON per year), dghs_pdfs/2023/, weather/, population/, download_manifest.csv
│   ├── parquet/       dghs_charts_long.parquet, weather_daily.parquet
│   └── processed/     national_daily, national_weekly, division_weekly, age_sex (parquet), splits.csv, cleaning_log.csv
├── notebooks/         01 to 05 (.ipynb)
├── r/                 06 to 09 (.R)
├── outputs/
│   ├── figures/       figures of notebooks 02, 04 and 05
│   ├── tables/        tables and findings of notebooks 02, 04 and 05
│   ├── sql_results/   q01 to q11 (.csv)
│   ├── models/        forecasts of every model (.csv), XGBoost models (.json)
│   ├── R/             one folder per script: 06_seasonality, 07_weather_dlnm, 08_statistical_forecasts, 09_model_comparison
│   └── powerbi/       47 tables, _relationships.csv, _table_dictionary.csv
├── Dengue Dashboard.pbix
└── Bangladesh Dengue Forecasting Analysis Report.pdf
```

Table 46. Run order.

Step	File	Runs in	Notes
1	notebooks/01_data_collection.ipynb	Google Colab	Downloads the dashboard, weather and population. Rerunning it refreshes the data and can change the numbers
2	notebooks/02_data_audit_cleaning.ipynb	Google Colab	Needs pdfplumber. Downloads the 2023 press releases on the first run
3	notebooks/03_duckdb_sql_analysis.ipynb	Google Colab	Eleven queries, one CSV each
4	notebooks/04_exploratory_analysis.ipynb	Google Colab	Figures and tables with the prefix 04
5	notebooks/05_ml_forecasts.ipynb	Google Colab	About six minutes. Writes the forecast file that script 08 reads
6	r/06_seasonality.R	RStudio	Reads data/processed
7	r/07_weather_dlnm.R	RStudio	Reads data/processed and data/parquet/weather_daily.parquet
8	r/08_statistical_forecasts.R	RStudio	A few minutes. Needs the output of notebook 05
9	r/09_model_comparison_powerbi.R	RStudio	Needs the outputs of all earlier steps. Writes outputs/powerbi
10	Dengue Dashboard.pbix	Power BI Desktop	Reads the CSV files in outputs/powerbi

Every notebook and script prints its findings from the data of the run and saves them as a text file, so the numbers quoted in this report can be traced to `04_findings.txt`, `05_findings.txt` and the `findings.txt` file of each R script.

B Software

Table 47. Software used.

Part	Software and settings
Python	pandas 2.2.3, scikit-learn 1.6.1, xgboost 3.4.1 (versions printed by notebook 05); numpy, duckdb, statsmodels, matplotlib, requests, pdfplumber, pyarrow
R	arrow, dplyr, tidyr, ggplot2, scales, splines, dlnm, MASS, forecast, urca
Dashboard	Power BI Desktop, on the 47 CSV tables in outputs/powerbi
Random seeds	random_state = 1 for XGBoost and the random forest. The R models are deterministic
SHAP	XGBoost's built-in TreeSHAP contributions (pred_contribs), median model

C Glossary of measures

These definitions are the ones shown in the dashboard tooltip.

Table 48. Measures used in the report and the dashboard.

Term	What it means	How it is calculated	How to read it
Admissions	People admitted to hospital with dengue in a week	DGHS daily counts added up over Sunday to Saturday	The main measure of burden. It counts hospital cases only, not all infections
Admissions per 100,000	Admissions for every 100,000 people living in the area	Admissions divided by the 2022 census population, times 100,000	Lets divisions of different size be compared fairly
Deaths per 100 admissions	How many admitted patients died, per 100	Deaths divided by admissions, times 100	A death rate among hospital cases, not among all infections
Epi week	Week number inside the epidemiological year	Weeks run Sunday to Saturday and belong to the year of their Thursday	Week 1 is the first week of January, week 52 the last of December
Adjusted week	A division-week whose value was corrected	Division values rescaled to agree with the national total, or rebuilt from press releases in 2023	An estimate, not a raw report. Marked in the weekly table
Season onset	The week the dengue season starts	First of three weeks in a row at or above 1,000 admissions nationally, or the same rate per person in a division	Depends on the threshold. Only 2024 is firmly later than the other years
Middle of the season	The week by which half of the year's admissions had arrived	Admissions added up through the year until they reach 50% of the year's total	A timing measure that needs no threshold. Complete years only
Seasonal strength	How much of the series is explained by the yearly cycle	From an STL decomposition: 1 minus remainder variance over seasonal plus remainder variance	0 = no yearly cycle, 1 = the series is all cycle
Relative risk	How many times the admissions, against typical weather	Negative binomial model with weather effects spread over lags of 0 to 12 weeks	1 = no effect, 0.5 = half, 2 = double. Clear only if the interval does not include 1
Lag	Weeks between the weather and the admissions it affects	Estimated by the weather model for each week from 0 to 12	The delay between a weather change and its effect on hospital cases
Origin week	The week in which a forecast is made	Each forecast uses only data up to and including this week	Forecasts for the same target week made in different origin weeks differ
Horizon	How far ahead a forecast looks	Target week minus origin week, 1 to 4 weeks	Errors grow with the horizon
Validation and test year	The two years the forecasts are scored on	2025 was used to choose features and settings of the machine learning models, 2026 was kept untouched	2026 is the fair score for the machine learning models. Both years are fair for the statistical models
Mean absolute error	The average miss of a forecast, in admissions	Average of the absolute difference between forecast and actual value, per division and week	Lower is better. Large divisions weigh more because their numbers are larger
Bias	Whether forecasts are too high or too low on average	Average of forecast minus actual	Negative = too low, the costly direction for a hospital
Better than last value	How much smaller the error is than that of repeating the latest week	1 minus the model's error over the last value's error, in %	Above 0 the model adds something. Below 0 it is worse than no model
80% range	The range a model expects the value to fall in 8 times out of 10	The 10th and 90th percentile of the model's forecast distribution	Honest if about 80% of actual values fall inside it
Weighted interval score	One score for the forecast and its range together	Half the absolute error plus a tenth of the range width and of ten times any miss outside the range, divided by 1.5	Lower is better. Rewards ranges that are narrow but still contain the value
Diebold-Mariano test	A test of whether two models' errors really differ	Compares the weekly errors of two models, allowing for neighbouring weeks being related	p below 0.05: the difference is unlikely to be luck
Ensemble	A combined forecast	The median of the five models' forecasts for each division, week and horizon	Usually more stable than any single model
SHAP share	How much a group of features drives the forecast	Average absolute SHAP contribution of the group over the total, per horizon	Shows what the model relies on, not what causes dengue